

**IMPLEMENTING SALIKSIKLAB: A RESEARCH MANAGEMENT
SYSTEM WITH SECURE MULTI-LANGUAGE CODE
EXECUTION AND DATA ANALYTICS FOR
SNSU CCIS DEPARTMENT**

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Bachelor of Science in Computer Science

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This thesis title “**Implementing SaliksikLab: A Research Management System with Secure Multi-Language Code Execution and Data Analytics for SNSU CCIS Department**”, prepared and submitted by **Vince B. Comon** and **Jeody Laine E. Saez**, in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science, has been examined and is hereby recommended for acceptance and approval for **ORAL EXAMINATION**.

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ABSTRACT

The increasing demand for efficient research management systems in academic institutions has highlighted the need for a centralized platform that can support research storage, retrieval, collaboration, and secure processing of academic outputs. This study developed SaliksikLab, a web-based research management system designed for the SNSU CCIS Department to address limitations in traditional repository systems such as poor organization, limited accessibility, and lack of integrated analytical tools. The main objective of the study was to design, develop, deploy, and evaluate a secure and scalable research repository system with role-based access control, multi-language code execution, and data analytics capabilities. The system was developed using a descriptive-developmental research design and followed an Agile Software Development Life Cycle. It utilized React for the frontend, Django REST Framework for the backend, and PostgreSQL for data storage, along with Docker and Nginx for deployment. Evaluation results showed that the system achieved a grand mean of 4.64, interpreted as Strongly Agree with a Very High qualitative description across functional suitability, performance efficiency, security, reliability, usability, and user satisfaction. The System Usability Scale also yielded a high overall mean of 3.97, indicating that users found the system easy to use, well-integrated, and suitable for frequent use. The findings confirmed that SaliksikLab effectively improves research organization, accessibility, and monitoring within the academic environment. Future studies are recommended to enhance scalability through improved system optimization and cloud-based deployment, expand the controlled code execution environment with additional programming language support, strengthen analytics features for deeper research insights, improve security mechanisms including authentication and access control, and conduct broader usability testing across multiple institutions to further validate system effectiveness.

Keywords: research management system, web-based repository, Agile development, system usability, academic analytics

TABLE OF CONTENTS

APPROVAL SHEET	ii
ACKNOWLEDGEMENT	iii
ABSTRACT	iv
TABLE OF CONTENTS	v
LIST OF FIGURES	vi
LISTS OF TABLES	vii
Chapter 1	1
1.1. Background of the Study	1
1.2. Statement of the Problem	3
1.3. Objectives of the Study	3
1.4. Significance of the Study	4
1.5. Scope and Limitation	5
1.6 Definition of Terms	8
Chapter 2	9
2.1. Local and Foreign Studies/Literature	9
2.2. Synthesis of the Review	17
Chapter 3	19
3.1. Theoretical Framework	19
3.2. Conceptual Framework	22
3.3. System Specification	23
Chapter 4	32
4.1. Research Design	32
4.2. Research Respondents	32
4.5. Research Deployment (System Testing, User Acceptance)	40
Chapter 5	42
5.1. Develop a Web-Based Research Repository Management System	42
5.2. Implement Multi-language Code Execution Environment	44
5.3. Integrate Data Analytics and Reporting Features	46
Chapter 6	56
6.1. Conclusion	56
6.2. Recommendations	57
REFERENCES	58
APPENDICES	62

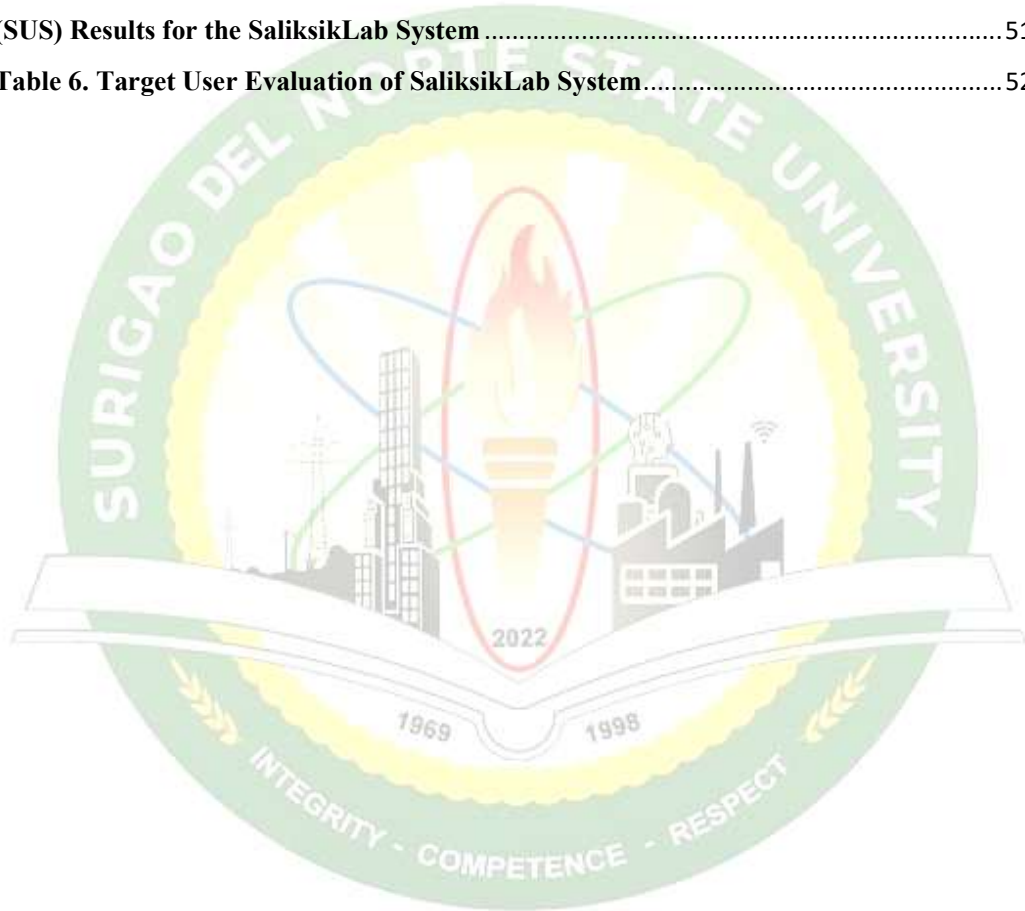
LIST OF FIGURES

Figure 1. IPO Model of the Study	22
Figure 2. Use Case Diagram for SaliksikLab System	29
Figure 3. Low-level System Design of SaliksikLab System	30
Figure 4. Agile Software Developmental Model of SaliksikLab System	35
Figure 5. Research Repository Management Interface	43
Figure 6. Multi-language Code Execution Environment Interface	45
Figure 7. Data Analytics and Dashboard Interface	47



LISTS OF TABLES

Table 1. Development Tech Stack	24
Table 2. Software Requirements of SaliksikLab System.....	26
Table 3. Security and Data Privacy Measure of SaliksikLab	27
Table 4. Demographic Profile of the Research Questionnaire Respondents by Sex, Course, User Role, System Experience, and Research Access Frequency During the Testing Phase of SaliksikLab	49
Table 5. Mean, Standard Deviation, and Interpretation of the System Usability Scale (SUS) Results for the SaliksikLab System	51
Table 6. Target User Evaluation of SaliksikLab System.....	52



Chapter 1

INTRODUCTION

1.1. Background of the Study

Research management systems have become essential tools in academic institutions for organizing, preserving, and utilizing research outputs in a more efficient and accessible manner. The study entitled “Implementing SaliksikLab: A Research Management System with Secure Multi-Language Code Execution and Data Analytics for SNSU CCIS Department” focuses on the development of a web-based platform that allows secure storage, management, execution, and monitoring of research-related systems and documents. The proposed system aims to provide a centralized repository where research papers, software projects, and executable systems can be uploaded and managed within a single environment. In addition, the system integrates a secure multi-language code execution environment that enables users to run different programming-based research systems remotely through tunneling technologies. The study also incorporates data analytics and reporting features to assist the department in monitoring research activities, system usage, and repository performance. Through these functionalities, the system promotes accessibility, efficiency, and sustainability of academic research outputs within the institution. Furthermore, the study addresses the common issue of valuable research systems being abandoned, unused, or stored as inactive files by providing a platform where these outputs can continuously be utilized for academic, instructional, and developmental purposes.

Several studies emphasized the importance of repository systems, research data management, and software quality evaluation in academic environments. According to Jaramillo et al., repositories for academic documents, algorithms, and datasets help institutions preserve research resources while improving accessibility and collaboration among researchers [1]. Goben and Sandusky discussed that open data repositories play a significant role in ensuring long-term research preservation; however, they also emphasized the growing concerns regarding security, long-term maintenance, and effective management of uploaded resources [2]. Similarly, He and Fang stated that institutional repositories support research data management by providing organized storage, metadata management, and controlled access mechanisms for academic outputs [3]. In terms of software evaluation, Canlas, Piad, and Lagman highlighted that systems intended for academic research management should undergo software quality

assessment using ISO/IEC 25010 standards to ensure functionality, usability, efficiency, and user satisfaction [4]. These studies collectively demonstrate the growing importance of research repository platforms and quality-driven systems in supporting institutional research management. Moreover, the findings of these studies serve as strong foundations for the development of SaliksikLab as a comprehensive research management platform for the SNSU CCIS Department.

Despite the continuous advancement of repository and research management systems, several limitations still exist in both local and global implementations. Many existing repository platforms primarily focus on storing research documents and datasets but lack support for executable research systems, multi-language code execution, and integrated analytics features. As a result, research outputs that involve software applications and system implementations are often archived as static files without long-term usability or practical accessibility. This limitation reduces the potential value of student-developed and faculty-developed systems that could otherwise be reused, enhanced, or referenced for future academic and institutional initiatives. Furthermore, some systems do not provide centralized monitoring and reporting tools that can help departments evaluate research productivity and repository usage effectively. Security and accessibility also remain concerns in hosting executable systems, particularly in environments where deployment resources are limited. These limitations highlight the need for a more comprehensive research management platform capable of supporting not only document preservation but also executable research outputs, analytics-driven monitoring, and secure accessibility. Addressing these concerns is important in maximizing the long-term usability, sustainability, and institutional value of academic research systems.

In response to these challenges, this study proposes the development of SaliksikLab, a web-based research management system that combines centralized repository management, secure multi-language code execution, and data analytics within a single academic platform. This system innovation offers a more sustainable approach to preserving and maximizing the usability of institutional research outputs by transforming them into accessible and functional digital resources. According to Matcha et al., learning analytics and monitoring systems provide significant support for improving decision-making, performance tracking, and institutional resource

management through data-driven insights [5]. Through the integration of secure multi-language code execution, the platform can support various programming-based research projects while enabling remote accessibility through tunneling technologies. Likewise, the incorporation of analytics and reporting features can assist administrators and researchers in monitoring repository activities, research trends, and system performance for more informed decision-making. The implementation of software quality evaluation standards can also guide the system toward achieving acceptable levels of performance, usability, and reliability. Ultimately, the proposed system seeks to ensure better preservation, accessibility, and long-term relevance of research outputs while promoting a more innovative and technology-driven research environment within the SNSU CCIS Department.

1.2. Statement of the Problem

Many academic institutions still rely on conventional repository systems and fragmented storage methods that primarily preserve research documents as static files, resulting in limited accessibility, underutilization of executable research systems, and difficulties in monitoring research productivity and system performance. Globally, higher education institutions continue to experience increasing volumes of digital research outputs and software-based projects, yet many repositories remain limited to document archiving without integrated execution environments, analytics capabilities, and centralized monitoring mechanisms necessary for sustaining research usability and institutional innovation. To address these critical deficiencies, the study proposes the development of SaliksikLab, a web-based research management system that integrates secure multi-language code execution, centralized repository management, and analytics-driven monitoring to improve accessibility, sustainability, evaluation, and long-term utilization of academic research outputs.

1.3. Objectives of the Study

General Objective

This study aimed to develop and evaluate SaliksikLab, a web-based research management system with secure multi-language code execution and data analytics for improving research accessibility, usability, monitoring, and management within the SNSU CCIS Department.

Specific Objectives

Specifically, this system:

1. Develop a web-based research repository management system
2. Implement multi-language code execution environment
3. Integrate data analytics and reporting features
4. Evaluate the developed system in terms of functional suitability, usability, efficiency, and user satisfaction

1.4. Significance of the Study

This study aimed to improve the management, accessibility, and analysis of research outputs in the university. Some research outputs in many academic institutions, especially in the College of Computer and Information Sciences (CCIS), were not deployed or utilized because there was no centralized system to store and manage them. Students also experienced difficulties in searching for related studies, executable systems, and archived research projects due to scattered storage methods and limited accessibility of academic resources. The proposed system provided a secure, web-based repository with multi-language code execution, centralized research management, and data analytics features for monitoring repository activities and research utilization. This system ensured that research outputs were preserved, easily accessible, and organized for future use. This study was beneficial for:

University: It strengthens the institution's research management by creating a centralized digital repository, improving data integrity, and supporting institutional compliance with research documentation and data privacy standards.

Administrators: It allows efficient monitoring, retrieval, and management of research outputs and software projects, reducing the risks of data loss and duplication, and supporting decision-making regarding research resources.

Instructors: It facilitates supervision and guidance by providing easier access to student research outputs, automated translation of abstracts, and tools for evaluating software-related projects.

Students: The system offers a developmental environment to test and execute code in multiple programming languages, which improved learning and experimentation. It also provided a reliable platform to browse existing research

as reference, identify research gaps, and develop new projects based on organized and accessible data.

Future Researchers: This study provides a foundation for further research into improving research management systems through emerging technologies such as artificial intelligence, automated analysis, multilingual processing, and secure computational environments.

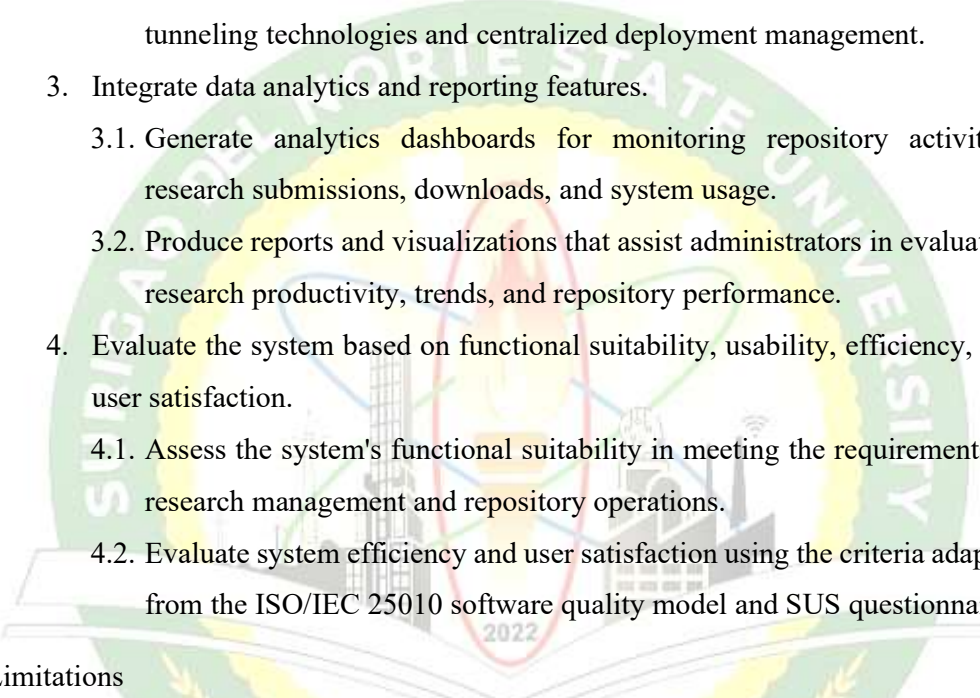
1.5. Scope and Limitation

The study was conducted at Surigao del Norte State University (SNSU), particularly within the College of Computing and Information Sciences (CCIS), from the second semester of Academic Year 2025–2026 to the first semester of Academic Year 2026–2027. The study focused on the design, development, deployment, and evaluation of SaliksikLab, a web-based research management system that provides centralized storage, organization, retrieval, and monitoring of academic research outputs. The system includes features such as research repository management, role-based access control, secure multi-language code execution, data analytics dashboards, research categorization, document archiving, and metadata-based search and filtering functions. The evaluation of the system was limited to selected students and instructors of the SNSU CCIS Department and was assessed using the System Usability Scale (SUS) and selected ISO/IEC 25010 software quality characteristics, specifically functional suitability, performance efficiency, security, reliability, and user satisfaction.

Scope

The core features of the research management system include:

1. Develop a web-based research repository management system.
 - 1.1. Provide centralized storage and management of research papers, executable system, datasets, and related academic documents.
 - 1.2. Implement role-based access control for students and faculty to ensure secure repository management
 - 1.3. Enable uploading, searching, retrieval, archiving, and categorization of research outputs based on title, author, keywords, year, and research abstract.

- 
- 1.4. Design responsive web design that accommodates computers, laptops, and mobile devices for improved accessibility.
 2. Implement a secure multi-language code execution environment.
 - 2.1. Support for executing research systems and programs developed using multiple programming languages within a controlled environment.
 - 2.2. Utilize containerized and isolated execution environments to minimize security risks during code execution.
 - 2.3. Provide remote accessibility of executable research systems through tunneling technologies and centralized deployment management.
 3. Integrate data analytics and reporting features.
 - 3.1. Generate analytics dashboards for monitoring repository activities, research submissions, downloads, and system usage.
 - 3.2. Produce reports and visualizations that assist administrators in evaluating research productivity, trends, and repository performance.
 4. Evaluate the system based on functional suitability, usability, efficiency, and user satisfaction.
 - 4.1. Assess the system's functional suitability in meeting the requirements of research management and repository operations.
 - 4.2. Evaluate system efficiency and user satisfaction using the criteria adapted from the ISO/IEC 25010 software quality model and SUS questionnaire.

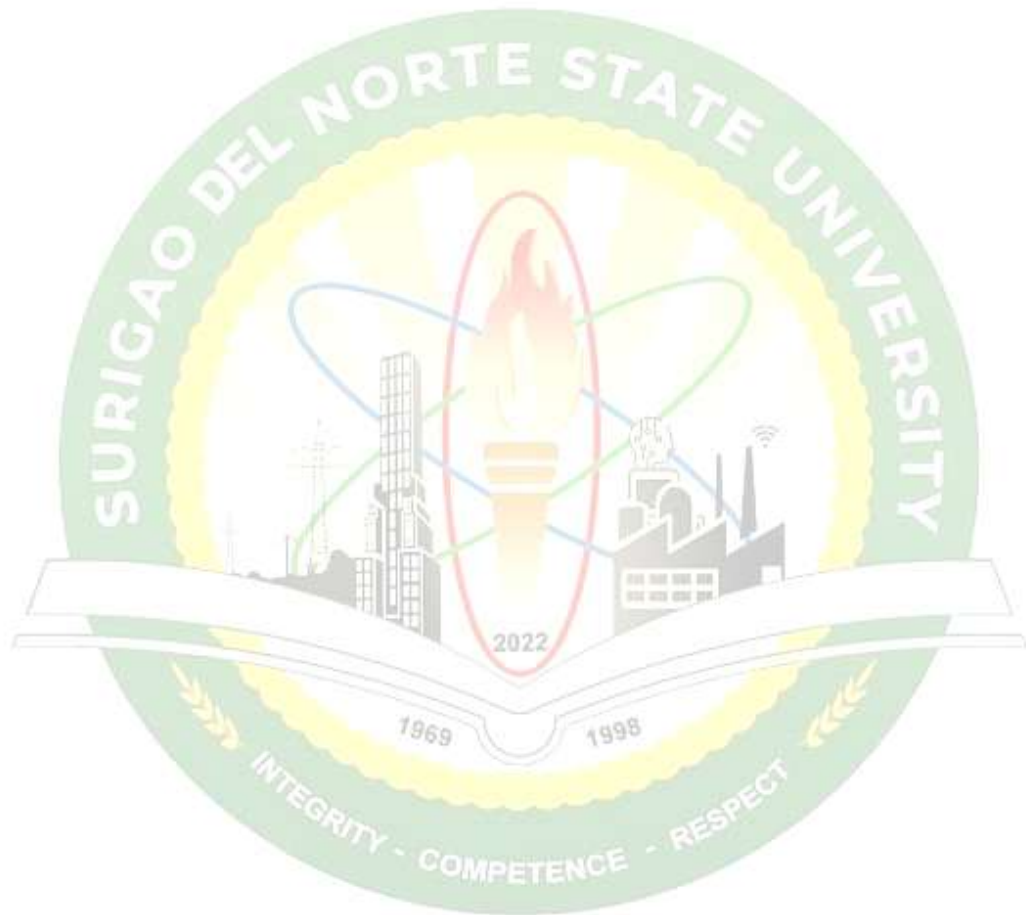
Limitations

However, there are certain limitations to the research management system, including:

1. The dependence on stable internet connectivity and tunneling services for remote access to deployed research applications, which may affect availability when network interruptions or tunneling service outages occur.
2. The security and performance constraints associated with executing user-submitted applications, as resource-intensive processes may require additional server capacity and monitoring mechanisms.
3. The system is limited to managing, storing, executing, and analyzing research outputs within the supported repository structure and programming environments. Research projects developed using unsupported technologies,

external dependencies, or specialized hardware requirements may not be fully executable within the platform.

4. The evaluation and deployment being limited to the CCIS Department of Surigao del Norte State University during the study period, which may require additional testing and validation before large-scale adoption in other academic units or institutions.



1.6 Definition of Terms

API. An application programming interface is a set of protocols and tools that allows the system to communicate with external applications and services.

Authentication. Verifying user identity before granting access to the system.

Automated Reporting. Generation of summaries, analytics, and logs from research outputs and code execution automatically.

Code Compilation. The process of converting source code into executable programs that can run within the secure sandbox environment.

Data Analytics. The process of collecting, processing, and analyzing of system data to generate insights regarding user activities, research productivity, trends, and repository performance.

Metadata. Descriptive details about research outputs, such as author, year, keywords, department, and adviser.

RBMC. A role-based management control is a method for managing user permissions and access based on assigned roles within the system.

Research Repository. A centralized digital platform for storing, organizing, and managing all institutional research outputs.

Runtime Testing. The execution of code in real-time to detect errors, validate output, and ensure program correctness.

Sandbox Environment. A secure and isolated space within the system where students can safely execute Python, Java, or C++ code without affecting other resources.

Tunneling Service. Create a secure communication channel between locally hosted research applications and external users, enabling multiple system to stored to be deployed, accessed, and executed remotely without requiring direct public hosting

Version Control. Tracking changes in research documents and software files to manage revisions.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1. Local and Foreign Studies/Literature

2.1.1 Research Repository Systems

Research management systems and institutional repositories are digital platforms designed to store, organize, and preserve academic research outputs in higher education institutions. These systems provide centralized access to scholarly works such as theses, dissertations, datasets, and research-based applications, ensuring that academic knowledge is properly managed and preserved over time. In modern academic environments, these platforms play a crucial role in improving the accessibility and discoverability of research outputs across different disciplines. They also support the organization of institutional knowledge by maintaining structured metadata and standardized documentation. As academic production continues to increase, the importance of efficient research management systems becomes more evident in ensuring sustainability and usability of research outputs. However, many existing systems remain limited to static document storage and lack support for interactive or executable research outputs. This limitation reduces the practical value of software-based academic projects that require execution and testing environments. As a result, there is a growing need for more advanced repository systems that go beyond traditional file storage functionalities.

Studies show that institutional repositories have significantly evolved as essential tools for managing academic outputs and improving research visibility across institutions. Jaramillo et al. emphasized that repositories for academic documents, algorithms, and datasets enhance educational utility by supporting structured access and long-term preservation of research materials [1]. Similarly, Bashir et al. highlighted that institutional repositories help reduce academic elitism by making research outputs more accessible to a wider academic community [2]. Liu et al. further explained that research management systems built on workflow-based architectures improve efficiency by automating submission, organization, and retrieval processes [6]. These systems demonstrate the importance of structured and automated research handling in academic environments. Additionally, Asadi et al. noted that institutional repositories are critical components in higher education institutions for managing scholarly outputs and

supporting knowledge dissemination [7]. Zibani et al. added that faculty research repositories often lack advanced integration features such as analytics and system interoperability, limiting their overall effectiveness [8]. Despite these advancements, most systems still focus primarily on document-based storage and do not fully support executable research systems or dynamic research environments.

Further analysis of existing systems reveals that research repositories in many institutions face challenges related to usability, accessibility, and system integration. Dube emphasized that institutional repositories are often used as passive storage systems for academic data rather than active platforms for research utilization [9]. Narayan and Luca also identified issues in the adoption of repository systems, including limited user engagement and challenges in integrating repositories into daily academic workflows [10]. Malaya et al. demonstrated that research information management systems can improve organization and accessibility in universities; however, many implementations remain isolated and lack scalability across departments [11]. Javier et al. further showed that institutional repository systems improve knowledge sharing but still require enhancements in usability and system integration for broader academic use [12]. Subiyakto et al. noted that usability evaluation is critical in repository system design, as poor interface design can significantly affect user adoption and satisfaction [4]. These findings collectively indicate that while institutional repositories are widely implemented, many systems still lack integration, usability optimization, and support for advanced functionalities. This gap highlights the need for a unified research management system that not only stores academic outputs but also enhances their usability, accessibility, and functional application within academic environments.

2.1.2 Centralized Research Repository and Knowledge Management Systems

Centralized research repository and knowledge management systems refer to integrated digital platforms designed to consolidate, organize, and manage academic research outputs within a unified environment. These systems aim to eliminate fragmentation of research data by providing a single access point for scholarly materials such as theses, datasets, publications, and software-based

outputs. In academic institutions, knowledge management systems play a critical role in ensuring that research outputs are properly stored, retrieved, and utilized for academic and institutional development. They also support efficient information flow by reducing redundancy and improving accessibility across departments. Furthermore, centralized systems enable structured data governance, allowing institutions to maintain consistency and integrity of research records. As the volume of academic data continues to grow, the need for unified research management infrastructures becomes increasingly important. However, many institutions still operate decentralized systems that limit collaboration and efficient knowledge sharing. This creates inefficiencies in research utilization and long-term academic value preservation.

Studies show that centralized research systems significantly improve the efficiency and usability of academic knowledge management in higher education institutions. Liu et al. emphasized that research management systems built on workflow-based architectures enhance automation, improve data organization, and support centralized control of research processes [6]. Similarly, Asadi et al. explained that institutional repositories play a key role in consolidating academic outputs and improving accessibility across universities by integrating scattered research data into a unified platform [7]. Jaramillo et al. further highlighted that repository systems designed for academic documents, algorithms, and datasets improve structured access and support knowledge sharing in educational environments [1]. These systems enhance research visibility while reducing duplication of efforts in academic institutions. Zibani et al. added that faculty research repositories contribute to knowledge preservation but often lack full integration and centralized coordination features [8]. Dube also noted that institutional repositories are increasingly viewed as reservoirs of research data, yet their potential is limited when systems remain isolated or fragmented [9]. Collectively, these studies indicate that while centralized systems exist, many still lack complete integration of research management functionalities across institutions.

Further analysis reveals that fragmentation remains a persistent issue in many academic knowledge management environments. Narayan and Luca identified challenges in institutional repository adoption, including lack of user

engagement and difficulties in integrating repositories into academic workflows [10]. Malaya et al. demonstrated that research information management systems can improve organizational efficiency, but many implementations remain limited to specific departments rather than institution-wide platforms [11]. Javier et al. further showed that repository systems enhance knowledge sharing but require stronger integration mechanisms to support scalability and centralized access [12]. Romero and Ventura emphasized the importance of data-driven academic systems, noting that learning analytics and structured data management improve institutional decision-making processes [3]. Jaramillo-Morillo et al. also highlighted that student interaction data extraction contributes to better understanding of system usage and research behavior in academic platforms [5]. However, despite these advancements, many systems still operate in fragmented environments where research outputs are not fully integrated or centrally managed. This reinforces the need for a unified research repository system that supports centralized storage, efficient knowledge management, and improved institutional research utilization.

2.1.3 Secure Multi-Language Code Execution Environments

Secure multi-language code execution systems refer to controlled computing environments that allow users to run programs written in different programming languages within a single platform. These systems are commonly implemented in online compilers, online judges, and cloud-based development environments that support programming education and software testing. A sandbox environment ensures that code execution occurs in an isolated space where system resources are protected from untrusted or potentially harmful code. This isolation is typically achieved through virtualization or containerization technologies that restrict access to the host system. Runtime constraints such as memory limits, CPU usage restrictions, and execution timeouts are applied to ensure system stability. These mechanisms are especially important in multi-user environments where multiple programs are executed simultaneously. Despite these safeguards, building a secure and scalable multi-language execution system remains complex due to differences in language runtime requirements and security risks. As academic institutions increasingly adopt digital platforms for programming education, the demand for secure and integrated execution environments continues to grow.

Secure execution environments have been widely studied in the context of sandboxing and isolated runtime systems. Malan introduced the CS50 sandbox system, which allows secure execution of untrusted code by enforcing strict isolation between the executed program and the host system [13]. This approach demonstrates how sandboxing can protect infrastructure while still enabling functional program execution. Rayhan et al. further emphasized that cybersecurity practices are essential in web-based applications that handle user-generated code, particularly in preventing injection attacks and unauthorized access [14]. In addition, containerization technologies such as Docker are commonly used to isolate execution environments, ensuring that each program runs independently without affecting system integrity. These approaches are often combined with API-driven architectures to manage execution requests securely and efficiently. However, most existing systems are designed for competitive programming or isolated software testing rather than integrated academic research execution environments. This creates a gap in systems that require both secure execution and academic workflow integration. As a result, multi-language sandbox systems must evolve toward more educational and research-oriented applications.

In many academic institutions, programming activities are still conducted using offline development environments such as local IDEs, which limit collaboration and remote execution capabilities. Students and researchers often rely on manual setup processes that do not support centralized execution or secure deployment. Malaya et al. noted that research information systems in academic environments tend to focus more on documentation and storage rather than executable system functionality [11]. Javier et al. also observed that institutional repositories primarily manage research outputs as static files, lacking support for runtime execution of software-based research projects [12]. Narayan and Luca further identified that limited infrastructure and usability challenges hinder the adoption of more advanced digital research platforms in academic institutions [10]. These limitations result in fragmented workflows where code execution, research storage, and system deployment are handled separately. Consequently, there is a lack of unified platforms that support both secure execution and academic research needs. This gap highlights the necessity for integrated sandbox systems that support multi-language execution within a controlled academic environment.

2.1.4 Remote Access, Tunneling, and Deployment Systems

Remote access and tunneling systems are technologies that enable secure communication between locally hosted applications and external users over the internet. These systems allow services running on private networks to be accessed remotely without requiring direct exposure to public IP addresses. Tunneling mechanisms such as reverse proxies and secure gateways create encrypted communication channels that forward traffic between clients and internal servers. Deployment architectures define how applications are hosted, accessed, and managed across distributed environments. In modern computing systems, these technologies are essential for enabling flexible, scalable, and secure application delivery. They are particularly useful in academic environments where infrastructure resources are limited. However, ensuring secure access while maintaining system performance remains a key challenge. As systems become more distributed, the importance of secure tunneling and remote deployment continues to increase.

Modern research emphasizes the role of secure gateways and architectural patterns in enabling scalable and protected remote access systems. Ochuba et al. explained that API gateway patterns provide a structured approach to managing traffic between clients and backend services while enforcing security policies [15]. These gateways act as intermediaries that regulate access and improve system scalability. Rayhan et al. also highlighted cybersecurity practices for web-based systems, emphasizing the importance of secure communication channels in preventing unauthorized access [14]. In addition, container-based deployment models allow applications to be exposed securely through tunneling services while maintaining isolation from core infrastructure. These architectures are often combined with Zero Trust Network Access principles to ensure that all access requests are continuously verified. However, despite these advancements, many systems remain complex to configure and maintain. Furthermore, most solutions are designed for enterprise or commercial environments rather than academic research systems. This creates a gap in simplified deployment solutions tailored for educational use.

In academic institutions, deployment and hosting infrastructure remain limited, especially in developing regions where server resources are scarce. Many

systems are deployed manually on local machines or basic hosting services without proper remote access configurations. This results in restricted accessibility, especially for students who need to access research systems outside institutional networks. Malaya et al. noted that academic information systems often lack scalable deployment strategies, limiting their usability and reach [11]. Javier et al. also found that most repository systems do not support dynamic deployment or remote execution capabilities, reducing their flexibility [12]. Additionally, the absence of tunneling technologies in academic environments forces institutions to rely on static hosting approaches. These limitations highlight the need for integrated systems that combine secure remote access, deployment automation, and centralized management. Therefore, tunneling-based architectures present a viable solution for enabling accessible and secure academic research systems.

2.1.5 Data Analytics and Research Monitoring

Data analytics in academic systems refers to the process of collecting, processing, and analyzing research-related data to generate meaningful insights for decision-making. These systems are used to monitor user activity, research submissions, system usage, and overall institutional productivity. In educational environments, data analytics plays a crucial role in improving performance tracking and identifying trends in research output. Dashboards are commonly used to visualize data in a structured and interpretable manner. These tools support administrators and researchers in making data-driven decisions. However, many traditional academic systems lack integrated analytics modules. This limits their ability to provide real-time insights into research activities. As a result, institutions often rely on manual reporting methods that are inefficient and time-consuming.

Research in learning analytics demonstrates the growing importance of data-driven systems in education. Romero and Ventura emphasized that educational data mining and learning analytics help institutions understand patterns in student and research behavior through computational analysis [3]. Jaramillo-Morillo et al. further explained that extracting interaction data from academic platforms enables institutions to monitor user engagement and system usage effectively [5]. These analytics systems typically include dashboards that track metrics such as usage frequency, submission rates, and performance indicators. Such data supports institutional planning and resource allocation. However, many

systems still operate in isolation without integrating analytics into repository or execution platforms. This separation limits the ability to correlate research activity with system performance. Therefore, integrated analytics systems are necessary to support comprehensive research monitoring.

In many academic institutions, particularly in developing regions, research monitoring is still performed manually through spreadsheets and periodic reports. This approach lacks real-time visibility and often leads to delayed decision-making. Subiyakto et al. noted that usability and system effectiveness are often overlooked in institutional systems, reducing their overall impact [4]. Zibani et al. also observed that faculty research repositories rarely include advanced analytics features for monitoring productivity [8]. These limitations make it difficult to evaluate research trends and system utilization accurately. As a result, institutions struggle to optimize research workflows and resource allocation. The absence of integrated analytics tools highlights the need for systems that combine repository management with real-time monitoring capabilities.

2.1.6 Software Quality Evaluation (ISO/IEC 25010 + SUS)

Software quality evaluation refers to the assessment of system performance, usability, reliability, and overall quality using standardized frameworks. One of the most widely used standards is ISO/IEC 25010, which defines key software quality characteristics such as functional suitability, performance efficiency, compatibility, usability, reliability, security, and maintainability. These attributes provide a structured way to evaluate whether a system meets user and institutional requirements. In addition to ISO standards, the System Usability Scale (SUS) is commonly used to measure user satisfaction and usability experience. SUS provides a quantitative score that reflects how users perceive system usability. These evaluation methods are widely adopted in academic and industrial software development. However, many educational systems still rely on informal evaluation approaches. This limits the reliability and comparability of system assessments.

Studies highlight the importance of standardized evaluation frameworks in ensuring software quality. Canlas et al. applied the ISO/IEC 25010 model in evaluating a faculty research productivity system, demonstrating its effectiveness in assessing functional and usability requirements [16]. Nurunnisa et al. also used ISO/IEC 25010 to evaluate web-based academic systems, confirming its relevance

in academic software assessment [17]. These studies show that structured evaluation frameworks provide objective measures of system quality. Brooke introduced the System Usability Scale as a reliable and widely used tool for evaluating user satisfaction [18]. Combined, ISO standards and SUS provide both technical and user-centered evaluation perspectives. However, many academic systems still do not adopt these standardized methods. This results in inconsistent evaluation outcomes across different studies and systems. Therefore, adopting formal evaluation frameworks is essential for ensuring system quality in academic environments.

In local academic environments, software evaluation is often conducted using simple surveys rather than standardized models. This approach limits the depth and accuracy of system assessment. Many student-developed systems are evaluated based only on user feedback without structured quality metrics. Subiyakto et al. highlighted that usability testing is often insufficiently applied in institutional systems, leading to incomplete evaluation results [4]. This indicates a gap in applying formal software quality models in academic research. Without standardized evaluation, it becomes difficult to compare system performance across studies. Therefore, integrating ISO/IEC 25010 and SUS into system evaluation processes is necessary to ensure consistent and reliable assessment of academic systems such as SaliksikLab.

2.2. Synthesis of the Review

There is limited literature available on the integration of research repository management, secure multi-language code execution, remote deployment, and analytics-driven monitoring within a single academic platform. Most studies focused on institutional repositories for storing and preserving research documents, improving knowledge sharing, and supporting research accessibility, but none addressed the combined management of executable research systems, secure sandbox-based code execution, and centralized deployment of software-based research outputs. Existing repository systems primarily function as document storage platforms and often lack mechanisms for executing, testing, and maintaining research applications beyond the archiving stage. Likewise, studies on sandbox environments and code execution platforms emphasize secure program execution and system isolation but are generally

designed for programming education, online judges, or software testing rather than academic research management. Research on tunneling technologies and remote deployment architectures mainly focuses on secure network access and application hosting, yet these solutions are rarely integrated with institutional research repositories. Furthermore, analytics-based studies highlight the value of monitoring user activities and research performance, but most implementations operate independently from repository and execution environments. Software quality evaluation studies also demonstrate the importance of ISO/IEC 25010 and System Usability Scale (SUS) frameworks in assessing system effectiveness and user satisfaction, although these standards have seldom been applied to comprehensive research management platforms that combine repository, execution, deployment, and analytics functionalities. These gaps indicate the absence of a unified academic system capable of preserving research outputs while simultaneously supporting secure execution, remote accessibility, centralized management, and data-driven monitoring. Hence, this study proposes the implementation of SaliksikLab, a Research Management System with Secure Multi-Language Code Execution and Data Analytics for the SNSU CCIS Department to provide a centralized, secure, and sustainable platform for managing, executing, monitoring, and preserving both research documents and software-based research outputs.



Chapter 3

THEORETICAL AND CONCEPTUAL FRAMEWORK

3.1. Theoretical Framework

Developing an intuitive web-based system for research management requires a strong theoretical foundation that explains how research outputs can be effectively stored, utilized, monitored, and sustained within an academic environment. This study is anchored on four major theoretical foundations: Knowledge Management Theory, Information Systems Success Model, Sandbox Security Theory, and Learning Analytics Theory. Knowledge Management Theory emphasizes the creation, organization, sharing, and utilization of knowledge resources to support organizational effectiveness and continuous learning [20]. In the context of higher education, research papers, datasets, source codes, and software projects represent valuable institutional knowledge that must be preserved and made accessible for future academic use. The theory suggests that centralized repositories enable institutions to transform scattered information into reusable knowledge assets that support innovation and collaboration. This perspective explains the need for SaliksikLab as a centralized platform where research outputs can be systematically organized, archived, and retrieved. The importance of repository systems in preserving and providing structured access to academic resources was demonstrated by Jaramillo et al., who emphasized the educational value of repositories for documents, algorithms, and datasets [1]. Similarly, Bashir et al. highlighted that institutional repositories improve accessibility to scholarly outputs and promote wider dissemination of knowledge within academic communities [2]. These concepts provide a theoretical foundation for developing an intuitive and user-centered research management system.

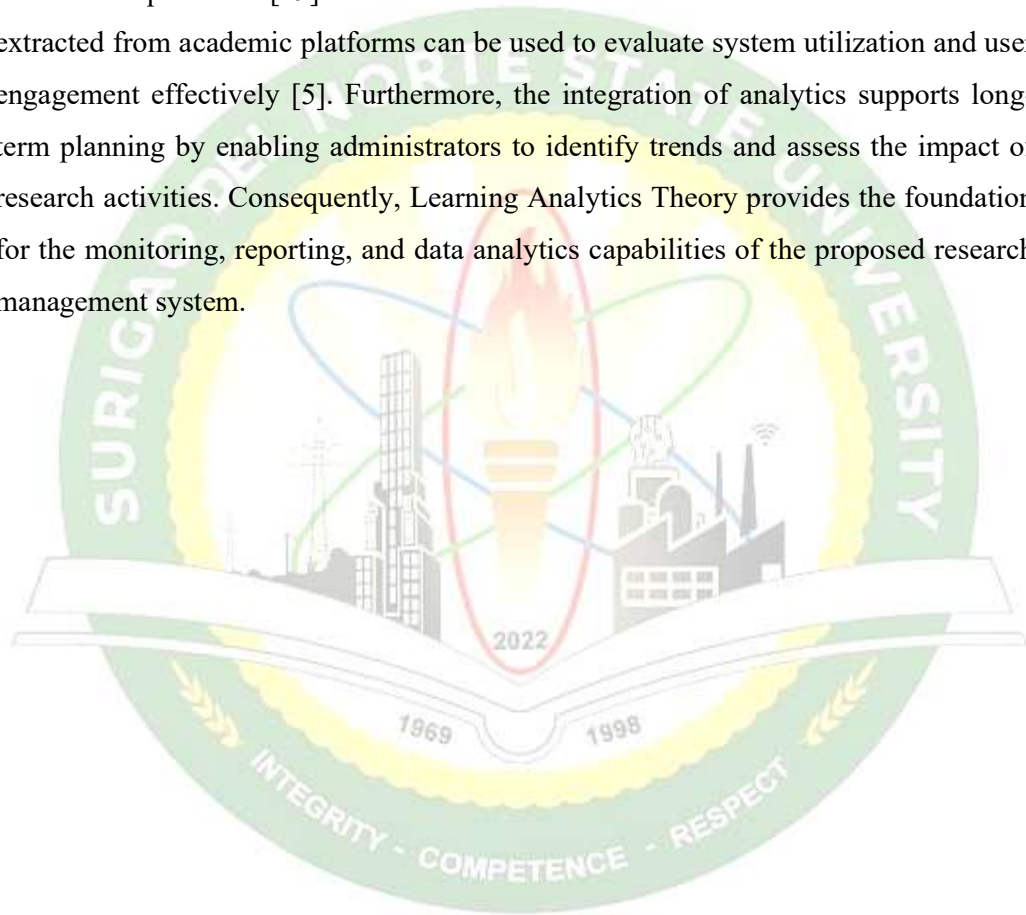
Another theoretical foundation of this study is the Information Systems Success Model, which describes how the quality of an information system influences user satisfaction, usage, and organizational benefits [21]. The model identifies key dimensions of system success, including system quality, information quality, service quality, user satisfaction, and net benefits. These dimensions are particularly relevant because SaliksikLab aims to provide an effective platform for managing and utilizing research outputs. The theory explains that users are more likely to adopt and continuously use a system when it is reliable, efficient, and capable of meeting their

needs. This principle supports the inclusion of usability, functionality, and performance considerations during system development and evaluation. Liu et al. demonstrated that workflow-based research management systems improve organizational efficiency through structured and automated management processes [6]. Subiyakto et al. further emphasized that usability is a critical factor affecting the acceptance and effectiveness of institutional repository systems [4]. The application of this model also justifies the evaluation of SaliksikLab using ISO/IEC 25010 quality characteristics and user-centered assessment methods. Consequently, the Information Systems Success Model provides the theoretical basis for assessing the effectiveness and quality of the proposed system.

The system architecture is also supported by Sandbox Security Theory, which explains the importance of isolating executable processes from critical system resources to prevent unauthorized access, malicious behavior, and system compromise. The theory asserts that untrusted programs should be executed within controlled environments where resource consumption and system interactions can be monitored and restricted. This concept is particularly important for SaliksikLab because the platform supports secure multi-language code execution for software-based research outputs. Through sandboxing and containerization mechanisms, user-submitted applications can operate independently without affecting the integrity of the host environment. The theory further supports the implementation of execution controls such as runtime limitations, resource allocation policies, and process isolation mechanisms. Malan demonstrated the effectiveness of sandbox-based architectures in securely executing untrusted code while maintaining system stability and security [13]. Rayhan et al. emphasized that secure execution environments and cybersecurity practices are essential for protecting web-based systems from unauthorized activities and vulnerabilities [14]. Ochuba et al. also explained that secure architectural patterns improve system scalability and controlled access to application services [15].

The fourth theoretical foundation is Learning Analytics Theory, which describes how data collected from user interactions and system activities can be analyzed to support informed decision-making and performance improvement [22]. The theory emphasizes the transformation of raw data into meaningful insights that help organizations monitor trends, evaluate performance, and optimize resources. Within the

context of this study, analytics functionality enables administrators to monitor repository utilization, research productivity, system activity, and user engagement. The theory explains that evidence-based decision-making becomes more effective when supported by timely and accurate analytical information. This principle justifies the inclusion of dashboards, reporting tools, and visualization features within SaliksikLab. Romero and Ventura highlighted that educational data mining and learning analytics provide valuable mechanisms for understanding patterns of behavior and improving institutional processes [19]. Jaramillo-Morillo et al. demonstrated that interaction data extracted from academic platforms can be used to evaluate system utilization and user engagement effectively [5]. Furthermore, the integration of analytics supports long-term planning by enabling administrators to identify trends and assess the impact of research activities. Consequently, Learning Analytics Theory provides the foundation for the monitoring, reporting, and data analytics capabilities of the proposed research management system.



3.2. Conceptual Framework

The conceptual framework of this study demonstrates how the input, process, and output (IPO) diagram works and how its components interact. The research process is structured around a conceptual model that outlines the flow of data and the development stages, as illustrated in the study's IPO Model (Figure 1).

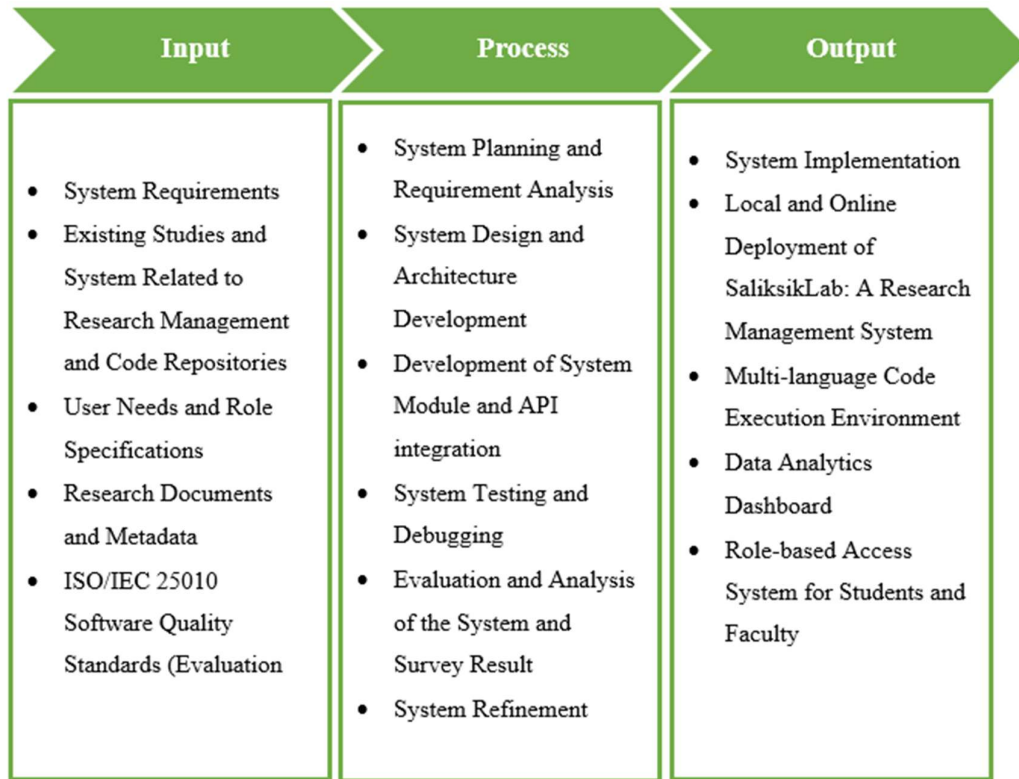


Figure 1. IPO Model of the Study

Figure 1 presents the conceptual framework based on the input-process-output (IPO) model, which shows how the study systematically moves the inputs to outputs. The input stage involves the gathering of system requirements in the form of informal interviews, observation, and documentary analysis and identification of the existing studies on research archive system and local problem on how to make undeployed system useful and still be accessible in academe. This also addresses the requirements of end users like student researchers and faculty and uses ISO/IEC 25010 quality standards as a guideline when developing and testing the system. These inputs also include research documents and metadata such as titles, authors, abstracts, keywords, and other relevant information needed for proper repository organization. Existing studies and systems related to research management and code repositories are also

considered to identify gaps and limitations in current implementations. The process stage involves system planning and requirement analysis, followed by system design and architecture development to ensure a structured foundation for the system. It also includes the development of system modules with API integration, system testing and debugging, and evaluation and analysis of system performance and survey results for improvement. Continuous system refinement is conducted to ensure that the platform meets functional and usability requirements based on identified standards. The output stage results in the implementation and deployment of SaliksikLab as a research management system with a multi-language code execution environment, data analytics dashboard, and role-based access system for students and faculty.

3.3. System Specification

This section describes the system's architecture, tools, and functional components.

3.3.1. System Architecture

The architecture follows a three-tier model consisting of the presentation layer, application logic layer, and data management layer:

1. **Frontend (Presentation Layer).** Developed using React with Vite to build a responsive single page application. It provides users with an interactive interface for accessing research repositories, analytics dashboards, authentication features, and system navigation across different roles and permissions.
2. **Backend (Application Logic Layer).** Utilized Django REST Framework that manages API endpoints and system business logic. It handles authentication, role based access control, repository management, analytics processing, and secure multi language code execution requests within a structured and scalable architecture.
3. **Database (Data Management Layer).** Built with PostgreSQL, which stores structured relational data including user accounts, research archives, metadata, logs, and analytics records. It ensures data integrity, efficient querying, and reliable storage of all system information used by SaliksikLab.

3.3.2 Development Tools and Technologies

The selection of modern frameworks and libraries used to build the system's infrastructure is summarized in the Development Tech Stack in Table 1:

Table 1. Development Tech Stack

Category	Tools/Technologies
Frontend	React 18, Vite, React Router, Axios, Chart.js, react-chartjs-2, Custom CSS
Backend	Django, Python, REST Framework
Database	PostgreSQL
Media Management	Django Media Files, MEDIA_ROOT, PDF upload storage, avatar upload storage
Authentication	Simple JWT, Django Custom User Model, Role-Based Access Control
Hosting and Deployment	Docker, Docker Compose, Nginx, Gunicorn, Cloudflare Tunnel, Cloudflare hostname support
Version Control	Git & GitHub

As shown in Table 1, the combination of these development tools enables the development of a scalable and secure web-based research management system for SaliksikLab. The frontend is built using React 18 with Vite which provides a fast and responsive user interface for browsing research repositories and accessing system features. React Router, Axios, and Chart.js support smooth navigation, efficient API communication, and interactive data visualization for analytics dashboards. The backend is developed using Django, Python, and Django REST Framework which handle core system logic including authentication, repository management, and analytics processing. PostgreSQL serves as the database which ensures reliable storage and retrieval of research documents, metadata, and user information. Media management tools such as Django Media Files and MEDIA_ROOT support secure uploading and storage of PDFs and user avatars within the system. Authentication is implemented using Simple JWT and role-based access control to ensure secure login and proper permission handling for students, faculty, and administrators. Hosting and

deployment tools such as Docker, Docker Compose, Nginx, and Gunicorn enable containerized and efficient system deployment while Cloudflare Tunnel provides secure remote access. Git and GitHub are used for version control to manage code changes and support collaborative development throughout the system lifecycle.

3.3.3 Functional Components

The system is composed of several functional components, each designed to handle specific research management, collaboration, and repository processes to ensure efficient and secure management of research documents and system code.

1. **User Authentication Module** – Handles user registration, login, logout, JWT token refresh, and password reset.
2. **User Profile Module** - Allows users to view and update their personal information and avatar.
3. **Role-Based Access Control Module** - Controls access to features based on user roles such as admin, faculty, and student.
4. **Account Approval Module** - Allows admins to approve or manage user accounts before system functionality access is granted.
5. **Research Management Module** – Manages the browsing, searching, filtering, viewing, previewing, and downloading of archive records.
6. **Archive Upload Module** – Allows users to upload academic PDF outputs with metadata such as title, author, course, department, keywords, and visibility.
7. **Archive Review Workflow Module** – Supports approval, rejection, and revision-request actions for submitted archive documents.
8. **Archive Versioning Module** – Tracks revised uploads and keeps version history for archive documents.
9. **Analytics and Dashboard Module** – Displays summary cards, archive activity, approval statistics, course distribution, and user engagement data.
10. **Admin User Management Module** – Allows admins to view users, change roles/status, approve accounts, and manage system users.
11. **Backup and Restore Module** – Allows admins to export CSV data and perform JSON backup or restore for repository records.

12. **File Management Module** – Handles secure file uploads, previews, downloads, and storage.
13. **Email Notification Module** - Supports password reset and account-related email sending through the configured email backend.

3.3.4 Software Requirements

The software environment necessary for the development and operation of the AI-based Research Management System is presented in Table 2.

Table 2. Software Requirements of SaliksikLab System

Software Component	Description
Windows 10 or higher	Used for system development, deployment, and administration tasks.
Google Chrome, Mozilla Firefox, Microsoft Edge	Provides cross-browser access to the web-based system.
Postman	Used for testing API endpoints and validating backend responses.
PostgreSQL	Stores user accounts, research files, metadata, and logs securely.
React and Python	Powers frontend interfaces and backend logic of SaliksikLab.
Visual Studio Code	Used for writing and managing system source code
Git	Version control system for source code management and collaboration.

The software requirements in Table 2 ensure that the AI-based research management system operates efficiently, reliably, and within a compatible development environment throughout its implementation and deployment phases. By using widely supported browsers such as Google Chrome and Mozilla Firefox, the system can be accessed without additional local installations and allows users to experience consistent performance, responsive interfaces, and cross-browser compatibility during research management tasks. The use of common programming languages and frameworks such as Python and React enables the system to deliver both

robust backend processing and an interactive frontend environment. Specifically, Python is utilized for backend services, API development, authentication, and research repository processing, while React is used to create dynamic, responsive, and user-friendly interfaces for document submission, repository browsing, and code execution modules. Additionally, the reliance on a standard database management tool such as PostgreSQL ensures that research files, system code, audit logs, user accounts, and other repository data are stored securely. Tools like Visual Studio Code and Git support development and collaboration by managing source code effectively. Postman plays a crucial role in testing and validating the Application Programming Interface (API) endpoints, ensuring that frontend and backend communication functions correctly. These software requirements establish a stable and scalable foundation for the SaliksikLab System.

3.3.5. Security and Data Privacy

The system employs specific protocols to safeguard user information and maintain data integrity, as summarized in Table 3.

Table 3. Security and Data Privacy Measure of SaliksikLab

Security Aspect	Description
User Authentication	SaliksikLab uses JWT-based authentication with login, token refresh, registration, password reset, and account approval controls.
Role-Based Access Control	Restricts system features and data access based on user roles using session information.
Data Encryption	Sensitive information are protected through HTTPS in deployment, while passwords are securely hashed by Django..
File and Input Validation	Archive uploads are limited to PDF files, and backend serializers validate submitted metadata and request data.

Code Execution Security	The system no longer includes AI model execution, collaboration execution features, or tunnel scripts, reducing exposure to unsafe code paths.
Secure Database Access	Limits database connections to authorized applications and uses parameterized queries to prevent unauthorized access.
Privacy Compliance	Public and private archive visibility settings help protect research files and user data according to access permissions

The security and data privacy measures presented in Table 3 are implemented that the system provides a secure, reliable, and privacy-compliant environment for managing academic research. The system uses JWT based authentication to control user access through login, token refresh, registration, password reset, and account approval mechanisms. This ensures that only verified users can access the platform and perform authorized actions within SaliksikLab. Role based access control is applied to restrict system features depending on user roles such as students, faculty, and administrators. Data encryption is enforced through HTTPS during deployment while passwords are securely hashed using Django security mechanisms to protect sensitive credentials. File and input validation is implemented to ensure that only valid PDF files are uploaded and that all metadata inputs are properly checked before storage. The system also applies secure database access by limiting connections to authorized backend services and using parameterized queries to prevent injection attacks. Privacy compliance is maintained through public and private visibility settings which allow users to control access to their research outputs. The code execution security aspect is strengthened by removing unsafe execution features and limiting exposure to risky processes within the system. These combined security measures ensure that SaliksikLab maintains confidentiality, integrity, and availability of research data. Overall, the implementation of these controls supports a safe and trusted research management environment for all users of the system.

3.3.6. Use Case Diagram

To define the functional boundaries and user roles within the systems, the interactions between students, instructors, and administrators are mapped in the use case diagram shown in Figure 2.



Figure 2. Use Case Diagram for SaliksikLab System

The diagram in Figure 2 illustrates how each actor interacts with the SaliksikLab System according to their assigned responsibilities. Students primarily engage with the system through various functionalities such as Upload Research Documents, Search and View Repositories, and Run Code Programs which enable them to submit their studies, access existing research resources, and execute system tools for academic purposes. In addition, students can browse within the system to assess existing research gaps that they may address in future studies, while the integrated AI title suggestion feature assists them in generating possible research titles and avoiding duplication of previously submitted works. Instructors are responsible for more evaluative functions, including Search and View Repositories, Run Code Programs, Approve Research Submission, and Review and Validate Research Submissions, which support the quality assurance and academic integrity of the repository. Meanwhile, administrators possess

the broadest access, which allows them to manage users, oversee repository content, monitor system activities, and generate repository reports for administrative and academic purposes. Each actor operates under a Role-Based Access Control (RBAC) structure, which restricts access to functions according to their responsibilities. The system boundary encompasses all core processes, indicating that all identified use cases are automated and managed within the SaliksikLab platform. Collectively, these interactions improve the efficiency of research management, strengthen submission validation workflows, and enhance collaboration among users within the academic research environment.

3.3.7. System Design

The technical architecture and modular breakdown of the platform, including the integration of external APIs, are detailed in the low-level system design in Figure 3.

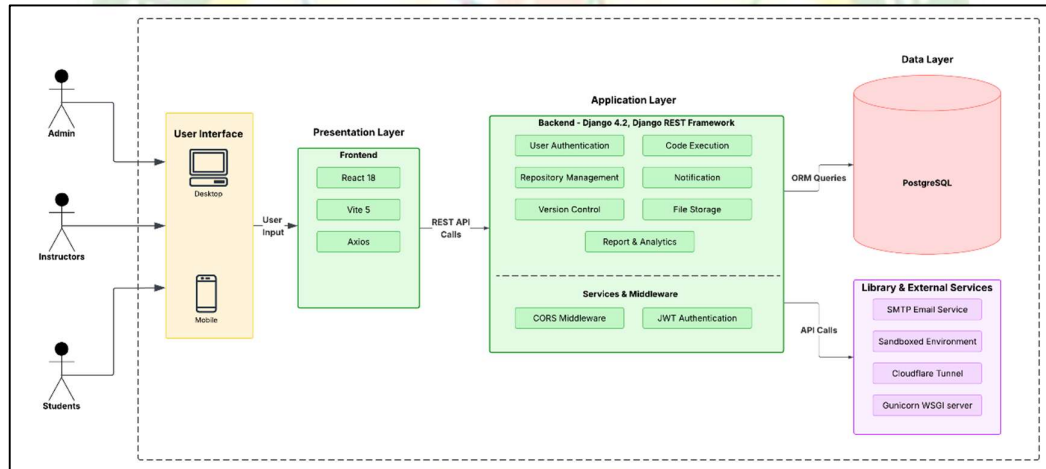
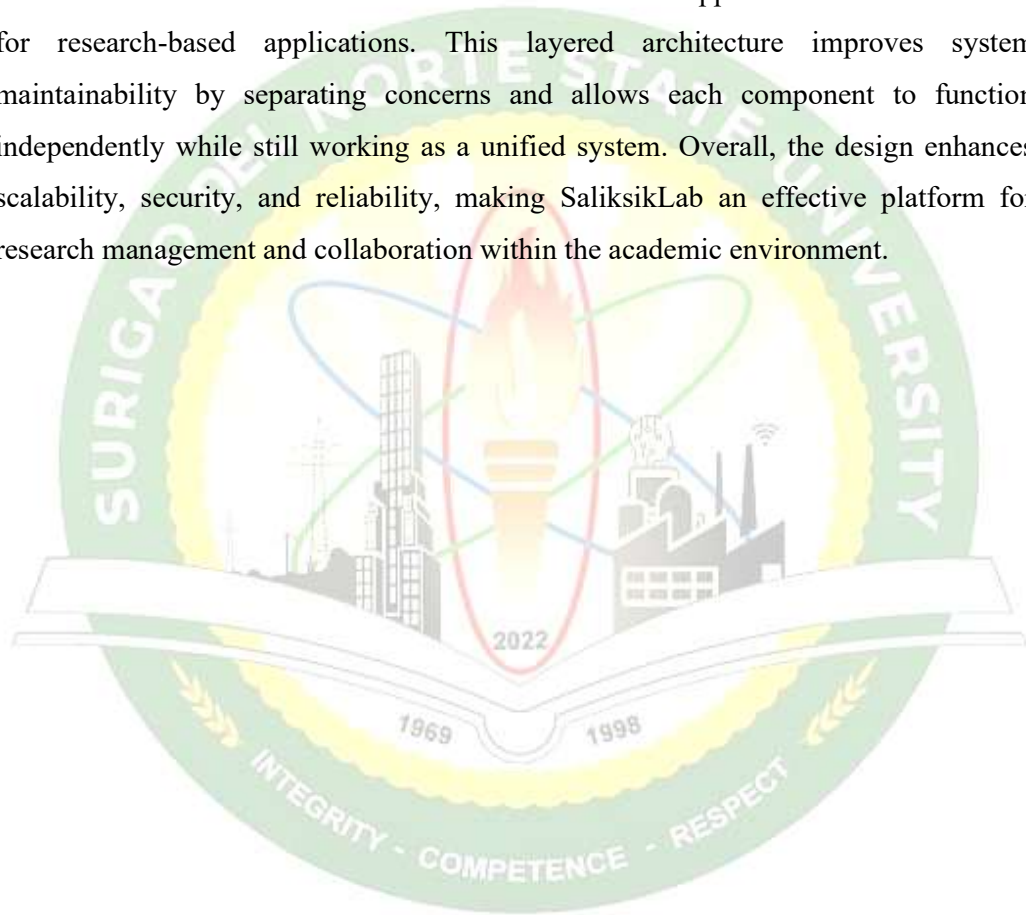


Figure 3. Low-level System Design of SaliksikLab System

According to Figure 3, the low-level system design of the SaliksikLab system illustrates the detailed technical architecture and modular composition that enable seamless research management and collaboration. The system is structured into three primary layers: the presentation layer, application logic layer, and data layer. The presentation layer, built with React 18, Vite 5, and Axios, provides an interactive and responsive user interface accessible on both desktop and mobile devices, ensuring that students, instructors, and administrators can efficiently navigate the platform. The application logic layer contains the core functional modules of the system, including

user authentication, research repository management, file handling, analytics processing, and notification services. It also incorporates middleware components such as JWT-based authentication and CORS configuration to ensure secure communication between the frontend and backend systems. The data layer is built using PostgreSQL and is responsible for storing all system information including user accounts, research outputs, metadata, logs, and activity records while maintaining data consistency and integrity. In addition, the system integrates external services such as SMTP for email notifications and a sandboxed execution environment to support secure code execution for research-based applications. This layered architecture improves system maintainability by separating concerns and allows each component to function independently while still working as a unified system. Overall, the design enhances scalability, security, and reliability, making SaliksikLab an effective platform for research management and collaboration within the academic environment.



Chapter 4

RESEARCH METHODOLOGY

4.1. Research Design

It specifically followed the principles of the Agile Software Development Life Cycle to support iterative development, continuous testing, and progressive refinement based on user feedback and evolving system requirements. This research design enabled the systematic planning and implementation of SaliksikLab while addressing the actual needs of students, faculty, and administrators in managing research outputs. The development phase involved designing a layered system architecture composed of a React-based presentation layer, a Django REST Framework application layer, and a PostgreSQL data layer. Core system modules were implemented, including user authentication, role-based access control, research repository management, file upload and metadata handling, analytics and reporting, and secure multi-language code execution within a sandboxed environment. Middleware services such as JWT authentication and CORS configuration were integrated to ensure secure and efficient communication between frontend and backend components. The system was deployed using Docker and Docker Compose, with Nginx serving the frontend and Gunicorn handling backend services, ensuring scalability, portability, and efficient performance across environments. Remote accessibility was supported through secure tunneling and hostname services to allow both local and online access to the platform. It was also made available locally for faculty and the CCIS Dean's Office to support on-site access and administration. To determine the effectiveness and reliability of the system, user evaluation was conducted among selected students and instructors who interacted with the platform during testing. The respondents assessed the system by answering the System Usability Scale questionnaire, while both quantitative and qualitative data were collected to evaluate the system's functional suitability, usability, efficiency, and user satisfaction. The findings from this evaluation provided actionable insights that guided system refinement, ensuring that SaliksikLab effectively addressed research management challenges and enhanced user experience within the university setting.

4.2. Research Respondents

The target respondents of the study are currently enrolled students at Surigao del Norte State University, particularly those in the College of Computing and

Information Sciences (CCIS), who participated in the SaliksikLab functionality demonstration during the pilot and evaluation phase. A purposive sampling technique was employed to ensure that the selected respondents were directly relevant to the study's objectives. This method allows the researchers to focus on individuals who can provide meaningful insights into the system's core workflows, accessibility, and overall efficiency. The sampling strategy also ensures representation across various year levels, academic backgrounds, and practical experience levels, which is important in evaluating how the system performs across different types of end users. Since SaliksikLab is intended to support research management, repository access, and code execution for academic purposes, selecting students from CCIS ensures that the respondents are familiar with digital systems, research processes, and collaborative development environments.

Inclusion criteria require that respondents must (1) be officially enrolled students at Surigao del Norte State University, (2) have completed or be currently participating in research-related courses, thesis development, or capstone project activities, and (3) be willing to actively engage with and provide feedback on the system. Meanwhile, exclusion criteria eliminate incomplete, duplicate, or invalid entries to maintain data reliability and integrity. For the qualitative component, only respondents who have actively used the system during the pilot phase are included to capture in-depth experiences and insights. Instructors are asked to evaluate both technical and pedagogical aspects of the system to identify potential gaps and opportunities for improvement. The sample size is determined to balance comprehensive feedback with feasibility of data collection and analysis. Respondents are assured of confidentiality and the voluntary nature of their participation to encourage honest and accurate responses. Finally, the combined perspectives of students, instructors, and administrators provide a holistic view of the system's effectiveness, usability, and potential for broader implementation.

4.3. Research Instruments / Tools

The study utilized several research instruments to collect data and feedback during the system testing and evaluation phases of SaliksikLab: A Research Management System. These instruments include survey questionnaires, structured interview guides, and system-generated logs. The survey questionnaire was adapted

from the System Usability Scale (SUS) and supplemented with standardized indicators aligned with ISO/IEC 25010 to evaluate usability, functional suitability, efficiency, and user satisfaction of the system. It was used to assess how students, instructors, and administrators perceived the overall performance and usability of the research repository system, including its analytics dashboard and secure code execution features. The structured interview guide was designed to gather qualitative feedback from users regarding their experience in navigating the repository, managing research outputs, accessing analytics, and interacting with role-based features. These interviews provided in-depth insights into system usability, accessibility, and the effectiveness of key functionalities such as authentication, repository workflows, and file management. System-generated logs were also utilized to capture objective data such as user authentication activities, file uploads, repository access, and system usage patterns, which helped validate actual user interactions within the platform.

In addition to evaluation instruments, the study employed several technical tools essential for system development and analysis. Unified Modeling Language (UML) diagrams were used during the design phase to illustrate system architecture, module relationships, data flow, and user interaction processes before implementation. The system was developed using React 18 and Vite for the frontend, providing a responsive and interactive user interface, while Django REST Framework and Python were used to implement backend services, APIs, authentication, and core business logic. PostgreSQL served as the central database management system for securely storing research documents, metadata, user accounts, and activity logs. Docker and Docker Compose were used for containerization and deployment to ensure consistency across development and production environments, while Nginx and Gunicorn handled frontend serving and backend request processing. Git and GitHub were used for version control to manage source code changes and support collaborative development. Additionally, built-in analytics and reporting modules within the system were used to continuously monitor research activities and system performance. Together, these instruments and tools ensured that both qualitative and quantitative data were effectively gathered, and that the system was properly developed, tested, and evaluated in a structured and reliable manner.

4.4 Research Implementation (SDLC Phases)

The development process of SaliksikLab system followed an Agile Software Developmental Model to ensure flexibility, iterative improvement, and continuous stakeholders' feedback throughout the project lifecycle. The model adopted in this study is illustrated in Figure 4

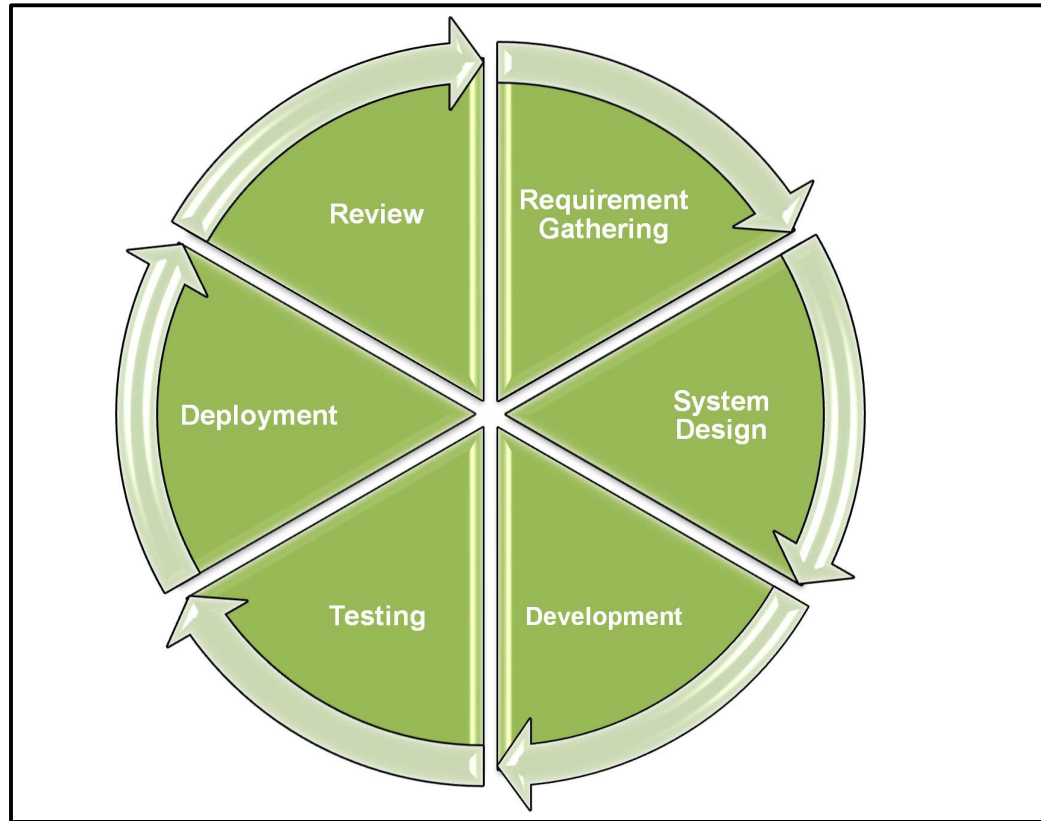


Figure 4. Agile Software Developmental Model of SaliksikLab System

As shown in Figure 4, the Agile Software Development Life Cycle (SDLC) was adopted in the implementation of SaliksikLab: A Research Management System to ensure iterative development, continuous improvement, and active user involvement throughout the development process. This approach allowed continuous feedback integration from students, instructors, and administrators to refine system features and align the platform with actual academic needs. During the planning and requirement analysis phase, the researchers identified key problems in existing research management practices such as fragmented storage of research outputs, lack of executable environments for software-based studies, and limited monitoring of research activities. These findings were used as the foundation for defining the system

requirements, including repository management, secure code execution, analytics, and role-based access control.

In the system design phase, the researchers developed the overall system architecture using a layered structure composed of a React-based presentation layer, a Django REST Framework application layer, and a PostgreSQL data layer. UML diagrams, database schema designs, and interface prototypes were created to guide structured implementation and ensure consistency across modules. The development phase focused on implementing core system functionalities including user authentication, research repository management, metadata handling, analytics dashboard, and secure multi-language code execution within a sandboxed environment using React, Django, and PostgreSQL. Middleware components such as JWT authentication and CORS configuration were also integrated to ensure secure communication between system layers.

After each development iteration or sprint, the testing and debugging phase was conducted to evaluate system functionality, detect errors, and ensure performance stability across different devices and browsers. Feedback from users during testing was incorporated into the refinement phase to improve usability, system responsiveness, security controls, and interface design. The deployment phase involved containerizing the system using Docker and Docker Compose, with Nginx serving the frontend and Gunicorn managing backend services, ensuring scalability and consistent performance across environments. The system was then made accessible both locally and remotely through secure deployment configurations. Overall, this iterative Agile-based implementation ensured that SaliksikLab evolved continuously based on user needs while maintaining reliability, security, and academic relevance.

3.4.1. Requirement Gathering

The requirement gathering phase involved collecting functional and non-functional requirements from students, instructors, and administrators to determine the core needs of the SaliksikLab: Research Management System. Data were gathered through interviews, observations, and informal discussions to identify existing challenges in research management such as fragmented storage of research outputs, limited accessibility of repositories, and the absence of a centralized system for

monitoring and managing academic research activities. From these activities, essential system features were identified, including research document upload and metadata management, repository browsing and search functionality, role-based access control, and analytics for monitoring research usage and productivity.

Stakeholders emphasized the need for a user-friendly web-based interface that allows users to efficiently upload, organize, and retrieve research outputs across different devices. Students and instructors also required secure access to research repositories with support for structured submission, revision tracking, and version management. In addition, the integration of a secure multi-language code execution environment using sandboxing mechanisms was identified as a key requirement to support software-based research outputs without requiring local software installation. Administrators and faculty members highlighted the importance of approval workflows, metadata validation, audit logs, and reporting features to ensure research quality control and proper academic supervision.

Security and data protection requirements were also prioritized, including JWT-based authentication, role-based access control, input and file validation, and secure database handling to ensure the confidentiality and integrity of research data. Non-functional requirements such as system scalability, fast response time, cross-device compatibility, and reliable deployment using containerized environments were also considered essential. Secure remote access through tunneling or hosted deployment was identified to support both local and online availability of the system. Overall, the gathered requirements ensured that SaliksikLab is aligned with institutional research processes and supports the development of a centralized, secure, and efficient research management platform for Surigao del Norte State University.

3.4.2. System Design

The system design phase of SaliksikLab focused on transforming the gathered requirements into a structured technical blueprint that guided system development. This phase involved the creation of Unified Modeling Language (UML) diagrams such as use case diagrams, class diagrams, activity diagrams, and system architecture diagrams to represent system processes, user interactions, and data flow. The system was designed using a layered architecture composed of a React-based presentation layer, a Django REST Framework application layer, and a PostgreSQL data layer to ensure modularity and scalability. Wireframes and user interface prototypes were developed

to define the layout of key modules such as the research repository, analytics dashboard, authentication system, and code execution environment. Database schema design was also performed using PostgreSQL to define structured relationships among users, research outputs, metadata, logs, and system activities. The design phase emphasized usability, security, and maintainability to ensure that the system supports efficient research management workflows. Overall, this phase served as the foundation for building a structured, scalable, and user-centered research management system.

3.4.3. Development

The development phase of SaliksikLab: Research Management System focused on converting the approved design into a fully functional web-based platform. The frontend was implemented using React 18 and Vite to create a responsive single-page application that supports seamless navigation across different user roles and devices. The backend was developed using Django and Django REST Framework to handle API services, authentication, repository management, analytics processing, and secure system logic. Core modules implemented during this phase include user authentication, role-based access control, research repository management, file upload and metadata handling, analytics dashboards, and secure multi-language code execution within a sandboxed environment. PostgreSQL was used as the primary database to store structured data such as user accounts, research outputs, metadata, and system logs. Middleware components such as JWT authentication and CORS configuration were integrated to ensure secure communication between frontend and backend systems. The development followed an Agile approach, allowing iterative implementation and continuous improvement based on testing results and stakeholder feedback. Security measures such as password hashing, input validation, and execution isolation were also implemented to ensure system integrity and reliability.

3.4.4. Testing

During the testing phase, SaliksikLab underwent systematic evaluation to ensure functionality, reliability, and usability of all system modules. Selected students and instructors participated in pilot testing by performing real-world tasks such as research submission, repository browsing, metadata searching, file uploads, and code execution within the sandbox environment. Functional testing was conducted to verify that all system features including authentication, repository management, analytics, notifications, and role-based access control worked as intended. Usability testing

assessed the interface design, navigation flow, responsiveness, and overall user experience across desktop and mobile devices. System logs and analytics dashboards were used to monitor performance indicators such as API response time, login success rates, file upload activity, and system usage patterns. Feedback was collected through surveys and interviews and was used to identify issues related to usability, workflow efficiency, and system responsiveness. Security testing was also performed to validate JWT authentication, access control rules, input validation, and sandbox isolation mechanisms. This iterative testing process ensured that the system met functional requirements and provided a stable and efficient platform for academic research management.

3.4.5. Deployment

The deployment phase involved making SaliksikLab fully operational and accessible to end users within and outside the university environment. The system was deployed using a containerized architecture with Docker and Docker Compose to ensure consistency across development and production environments. The frontend React application was served through Nginx, while the backend Django REST API was handled by Gunicorn for efficient request processing. PostgreSQL was deployed as the primary database system to manage all persistent data including user accounts, research outputs, and system logs. Secure access was implemented using HTTPS encryption and domain-based routing, with optional tunneling support for remote access. The system was also made available for local use within faculty offices and administrative units to support internal monitoring and approval workflows. This dual deployment approach ensured both online accessibility for users and controlled local access for institutional management. Overall, the deployment strategy ensured that SaliksikLab operates as a secure, scalable, and reliable research management system ready for academic use at Surigao del Norte State University.

3.4.6. Review

In the review phase, the researchers evaluated the overall effectiveness, usability, functionality, and performance of SaliksikLab: A Research Management System. Selected students, instructors, and administrators who participated in the pilot implementation were asked to assess the system using the System Usability Scale (SUS) and evaluation criteria adapted from the ISO/IEC 25010 software quality model. The evaluation focused on key system features such as research repository

management, role-based access control, file upload and retrieval, analytics dashboards, and secure multi-language code execution. Qualitative feedback was also collected through interviews and user observations to identify strengths, limitations, and areas for improvement. System-generated logs and analytics reports were analyzed to validate actual system usage, monitor repository activities, and assess overall system performance. Security mechanisms including authentication controls, access permissions, file validation, and data protection measures were also reviewed to ensure compliance with institutional requirements. The review phase enabled the researchers to identify functional and usability issues and implement necessary refinements to improve system quality. Overall, the evaluation provided valuable insights that supported the enhancement of SaliksikLab as a secure, reliable, and user-centered platform for managing academic research outputs within Surigao del Norte State University.

4.5. Research Deployment (System Testing, User Acceptance)

The deployment phase involved implementing SaliksikLab for actual use and evaluation by its intended users, including students, instructors, and administrators. During this phase, the system underwent comprehensive testing to verify the functionality, reliability, usability, and security of its major components, including user authentication, research repository management, metadata handling, analytics dashboards, role-based access control, and secure multi-language code execution. Pilot testing was conducted under realistic academic scenarios to evaluate how effectively users could perform research management tasks using the platform. Feedback was collected through the System Usability Scale (SUS), interviews, and direct observation of user interactions to identify potential issues and areas requiring improvement. User acceptance testing focused on determining whether the system met stakeholder expectations in terms of accessibility, ease of use, workflow efficiency, and overall functionality. System-generated logs and analytics data were also examined to monitor performance indicators such as login activity, repository usage, file upload success rates, and system responsiveness. Findings from the testing process were used to refine system features and improve the overall user experience. The combination of system testing and user acceptance evaluation ensured that SaliksikLab was ready for operational deployment and capable of supporting the research management needs of

the SNSU CCIS Department. Ultimately, this phase confirmed the system's effectiveness in providing a centralized, secure, and efficient platform for managing, preserving, monitoring, and utilizing academic research outputs.



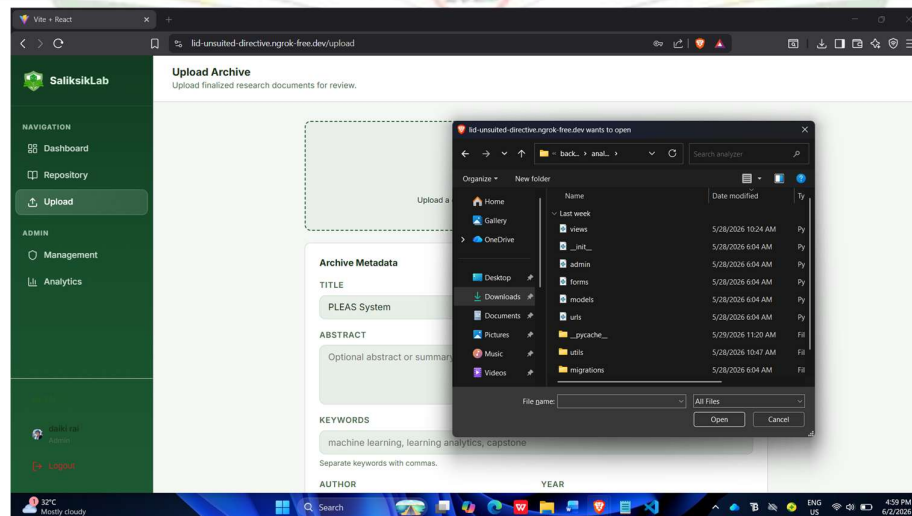
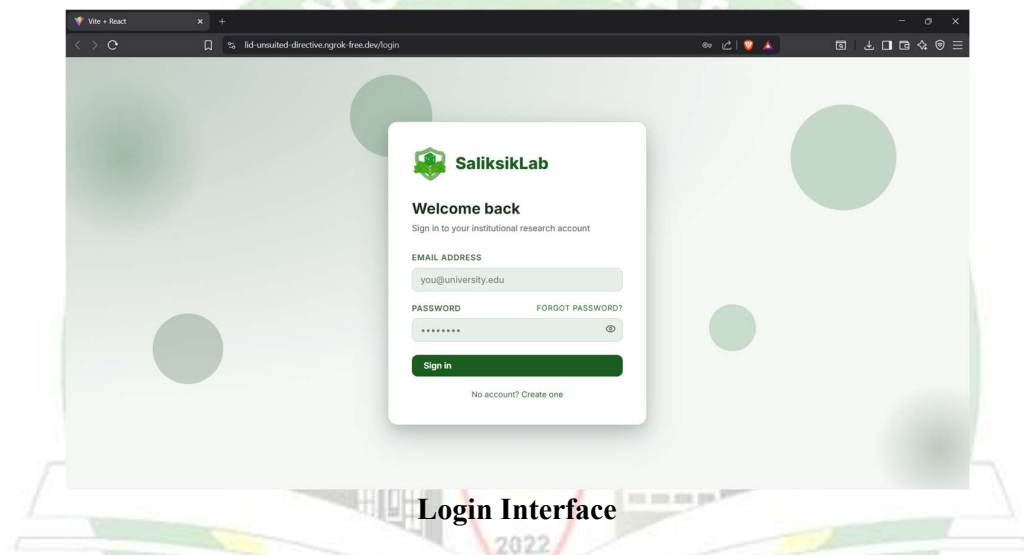
Chapter 5

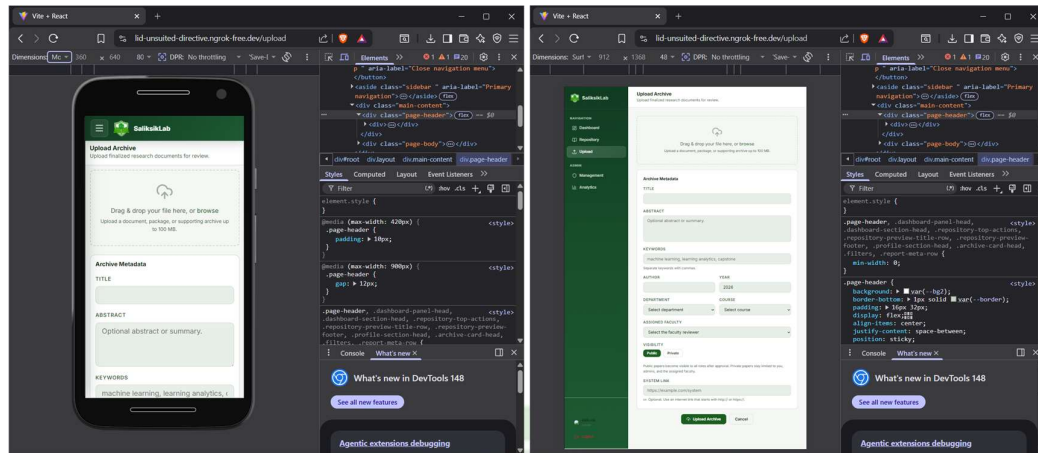
RESULTS AND DISCUSSIONS

This chapter presents the results and discussion of the developed SaliksikLab: A Research Management System for the SNSU CCIS Department. The presentation is organized based on the four specific objectives and their corresponding scopes.

5.1. Develop a Web-Based Research Repository Management System

The Research Repository Management System, as shown in Figure 5, presents the centralized interface used for storing, organizing, managing, and accessing research outputs within the SaliksikLab platform.





Responsive Web Design Compatible for Mobile Devices and Tablet

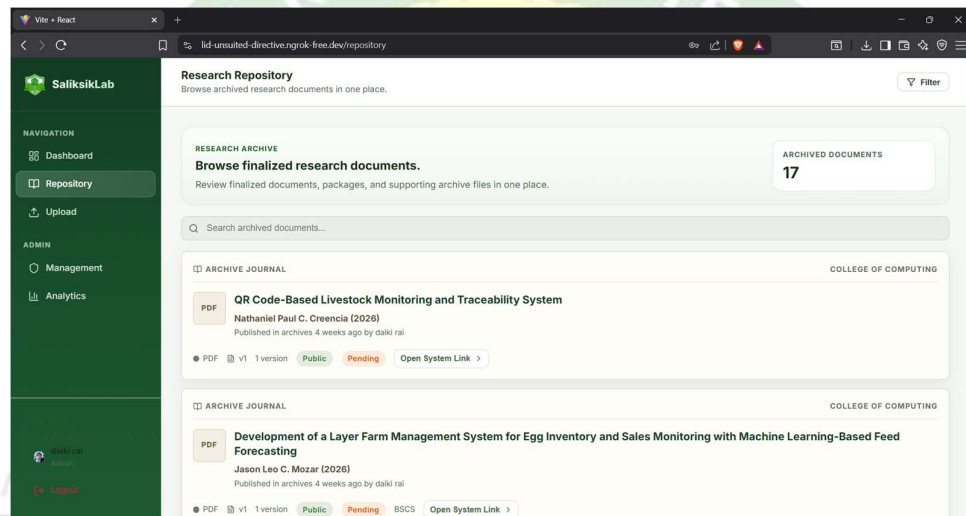


Figure 5. Research Repository Management Interface

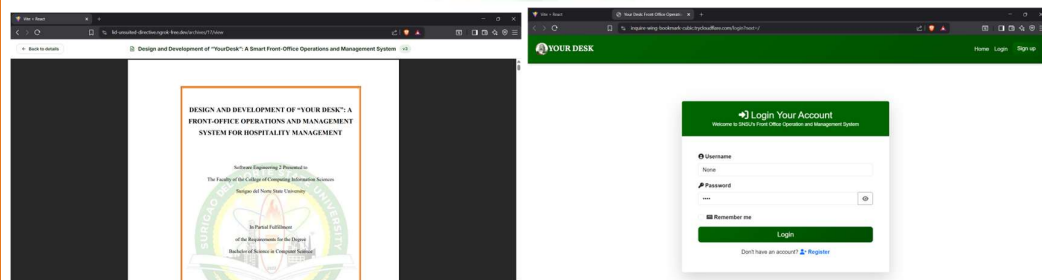
The Research Repository Management Interface serves as the core component of SaliksikLab by providing a centralized repository for academic research outputs and related resources. Through this interface, students and faculty can upload, manage, and preserve research papers, datasets, and software-based research projects within a single platform. The system eliminates fragmented storage practices by consolidating research materials into a structured repository that supports long-term preservation and accessibility. Role-based access control is implemented to ensure that users only access functions and resources appropriate to their assigned roles. Students are allowed to submit and manage their research outputs while faculty members can review, evaluate, and monitor repository contents. This approach strengthens repository security and promotes organized management of academic resources. The interface also provides clear navigation and management tools that simplify repository operations for users

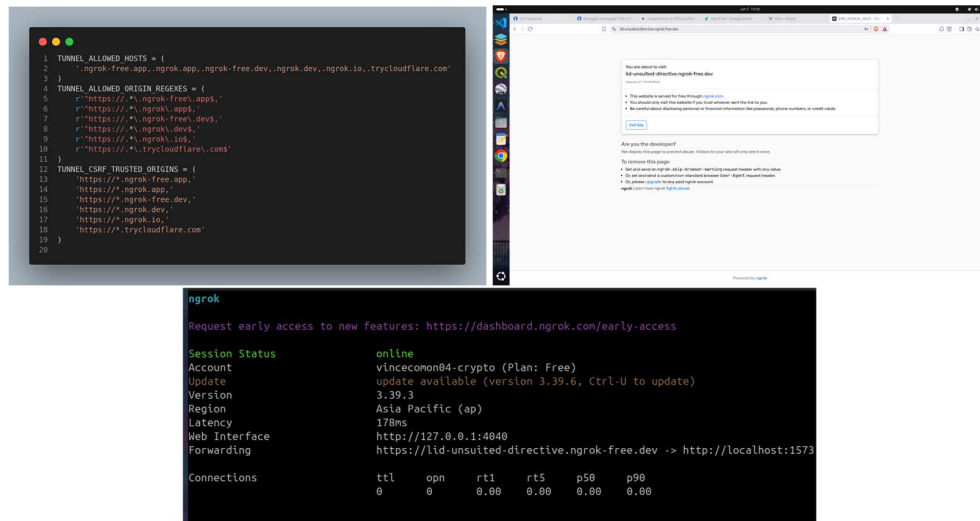
with varying levels of technical expertise. According to Jaramillo et al. [1], centralized repository systems improve the preservation, accessibility, and utilization of academic resources by providing structured storage and management mechanisms. The implementation of these capabilities demonstrates that the system successfully achieved the objectives of centralized storage and secure repository management.

Furthermore, the interface supports efficient uploading, searching, retrieval, archiving, and categorization of research outputs through metadata-based organization. Users can locate research materials using attributes such as title, author, keywords, publication year, and abstract, allowing faster access to relevant information. The repository also provides archiving functions that preserve completed research outputs for future reference and institutional use. To enhance accessibility, the system was developed using responsive web design principles that allow seamless operation across desktop computers, laptops, tablets, and mobile devices. This design ensures that users can access repository services regardless of their preferred device or location. The responsive layout automatically adjusts interface elements to maintain usability and readability across different screen sizes. Such flexibility supports wider adoption and encourages continuous utilization of the repository among students and faculty members. He and Fang [3] emphasized that institutional repositories enhance research management by providing organized metadata structures, efficient retrieval mechanisms, and controlled access to academic outputs.

5.2. Implement Multi-language Code Execution Environment

The Multi-language Code Execution Interface, as shown in Figure 6, presents the environment that allows users to execute and access software-based research projects developed using different programming languages within the SaliksikLab platform.





Tunnel Services

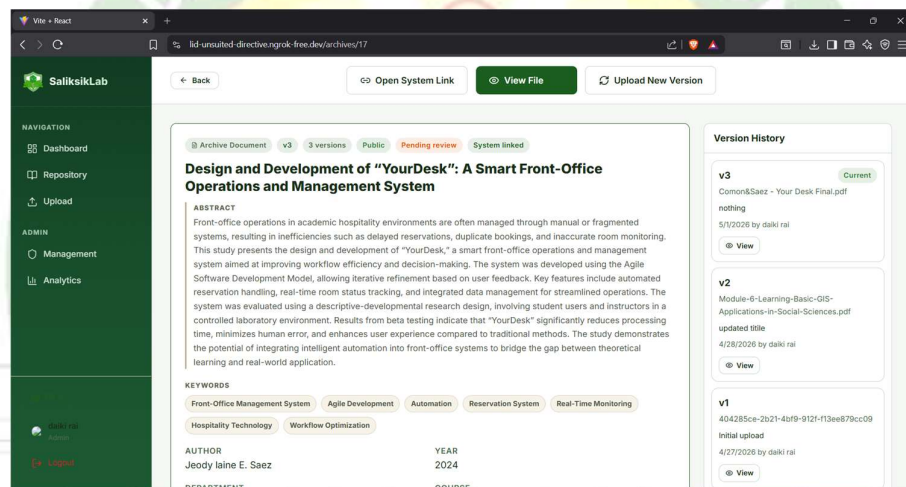


Figure 6. Multi-language Code Execution Environment Interface

The Multi-language Code Execution Interface was developed to support the execution of software-based research outputs within a secure and centralized academic environment. Through this interface, users can access and run research systems developed using various programming languages without requiring manual installation or configuration on their local devices. This capability allows students, instructors, and researchers to interact directly with executable research outputs rather than relying solely on static documentation and screenshots. The interface provides a unified platform where software projects can be deployed, managed, and accessed through a web-based environment. By supporting multiple programming technologies, the system accommodates the diverse software development requirements of research projects within the CCIS Department. The execution environment enables users to evaluate the

functionality and behavior of research systems in real time, which improves the usability and long-term value of software-based outputs. This feature addresses the common issue of research applications becoming inaccessible after project completion due to deployment limitations. Malan [13] emphasized that secure execution platforms enable the safe operation of user-submitted applications while preserving functionality and accessibility. Therefore, the developed interface successfully achieves the objective of supporting research systems developed using multiple programming languages within a controlled environment.

To ensure security and operational stability, the system utilizes containerized and isolated execution environments that separate running applications from the host infrastructure. Each executable research system operates within a controlled environment where resource usage and application behavior can be managed independently. This approach minimizes security risks by preventing applications from directly accessing critical system resources and other deployed projects. In addition, the platform supports remote accessibility through tunneling technologies and centralized deployment management. These technologies allow executable research systems to be accessed securely through the internet without requiring complex server configurations or public hosting services. As a result, deployed applications remain available to authorized users regardless of their physical location. The centralized deployment mechanism also simplifies system administration by providing a single platform for managing and monitoring research applications. Ochuba et al. [15] explained that secure gateway architectures and controlled access mechanisms improve the accessibility, scalability, and protection of remotely deployed applications. Consequently, the implementation of containerization, isolated execution, and tunneling technologies demonstrates that the system successfully fulfills the objectives of secure code execution, remote accessibility, and centralized deployment management.

5.3. Integrate Data Analytics and Reporting Features

The Data Analytics and Dashboard Interface, as shown in Figure 7, demonstrates the integrated analytics and reporting capabilities of the SaliksikLab system for monitoring repository activities, research outputs, and system utilization.

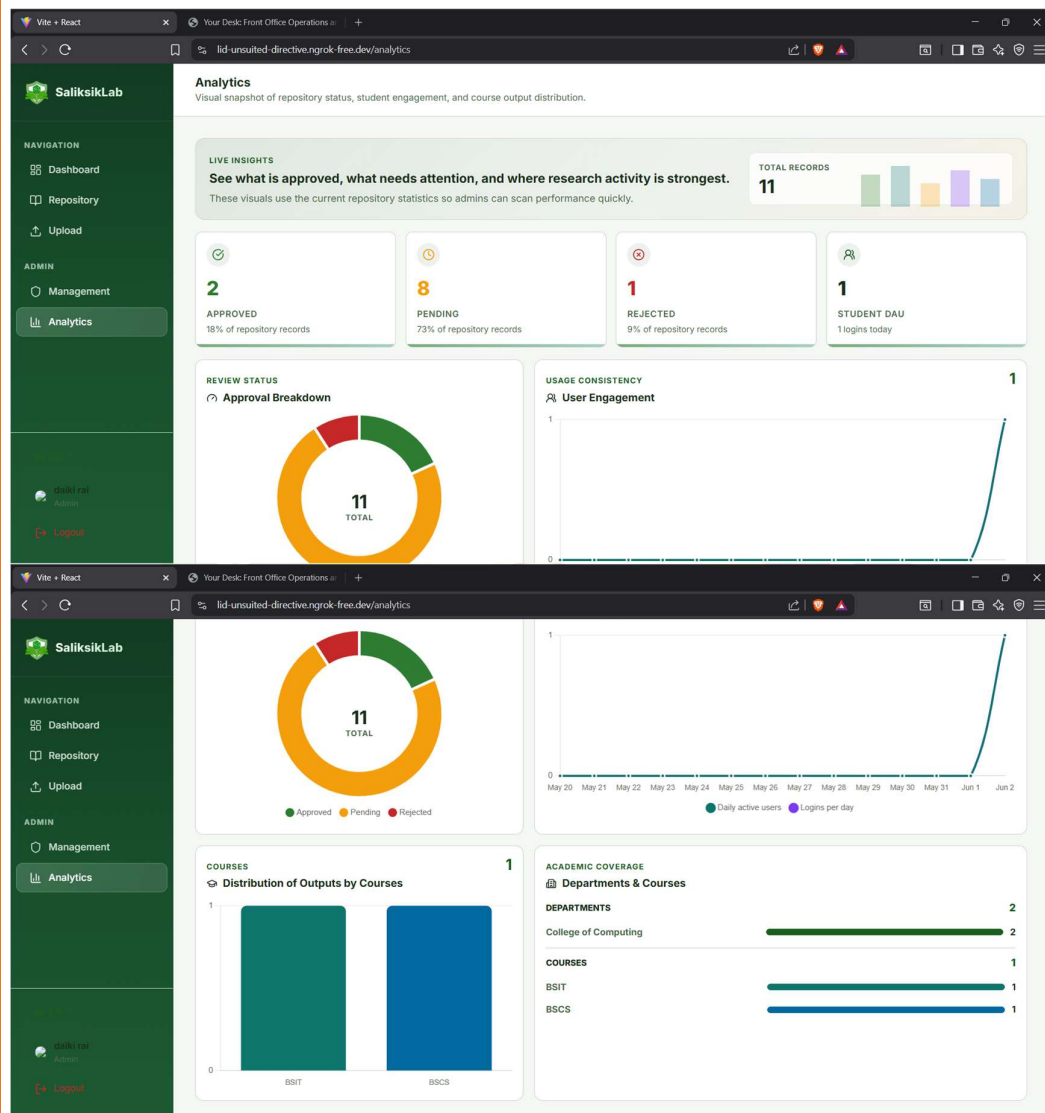


Figure 7. Data Analytics and Dashboard Interface

The Data Analytics and Dashboard Interface provides administrators and faculty members with a centralized view of repository performance and user engagement within the SaliksikLab system. Through the dashboard, users can monitor the number of research submissions, approved research outputs, archived documents, downloads, and overall repository activities in real time. The interface presents information through graphical visualizations and summary cards that make large volumes of data easier to interpret and analyze. Administrators can quickly identify trends in research production across different courses, departments, and academic years. The dashboard also supports monitoring of user participation by displaying

submission activities and repository usage statistics. These analytics help decision makers evaluate how effectively the repository is being utilized by students and faculty members. The integration of interactive charts and statistical summaries improves the accessibility of information and reduces the effort required to generate manual reports. According to Shneiderman et al. [23], effective visual analytics dashboards support better understanding of complex datasets by enabling users to identify patterns and trends through interactive visual representations. The successful implementation of these features demonstrates that the system achieved the objective of generating analytics dashboards for monitoring repository activities, research submissions, downloads, and system usage.

The reporting component further strengthens the capability of SaliksikLab by providing structured reports and visual summaries that assist administrators in evaluating research productivity and repository performance. Reports can be generated from the collected repository data and presented through charts, graphs, and statistical summaries that highlight significant research trends. The system enables administrators to examine the volume of research outputs produced over specific periods and determine which academic programs contribute the highest number of submissions. These reports support evidence-based decision making by providing accurate and timely information regarding research activities within the department. Faculty members and administrators can also use the generated reports to assess repository growth and identify opportunities for improving research engagement. The availability of automated reporting reduces administrative workload and increases the efficiency of monitoring institutional research outputs. Furthermore, the visual presentation of information enhances data interpretation and supports strategic planning for future academic initiatives. Research by Few [24] emphasized that dashboards and reporting systems improve organizational performance by transforming raw data into meaningful information that supports monitoring and decision making. Therefore, the developed analytics and reporting features successfully fulfilled the objective of assisting administrators in evaluating research productivity, identifying trends, and assessing overall repository performance.

5.4. Evaluate the System Based on Functional Suitability, Usability, Efficiency, and User Satisfaction

The system evaluation, as presented in this section, assesses the overall performance of SaliksikLab in terms of functional suitability, usability, efficiency, and user satisfaction based on user feedback and standardized evaluation criteria.

Table 4. Demographic Profile of the Research Questionnaire Respondents by Sex, Course, User Role, System Experience, and Research Access Frequency During the Testing Phase of SaliksikLab

Profile	Frequency (f)	Percent (%)
Sex		
Male	62	46.27
Female	72	53.73
Total	134	100.00
User Role		
Student	124	92.54
Instructor	10	7.46
Total	134	100.00
Course		
BSCS	52	38.81
BSICT	41	30.60
BSIS	41	30.60
Total	134	100.00
Have you used a research repository system before?		
Yes	89	66.42
No	45	33.58
Total	134	100.00
How often do you access research materials or repositories?		
Daily	34	25.37
Weekly	56	41.79
Monthly	28	20.90
Rarely	16	11.94
Total	134	100.00

The table 4 shows that the demographic profile of the 134 SUS respondents during the testing phase of SaliksikLab is composed of a slightly higher number of female participants at 53.73 percent compared to male participants at 46.27 percent, indicating a relatively balanced gender distribution with a mild female dominance. In

terms of user role, the majority of respondents are students at 92.54 percent while instructors represent only 7.46 percent, which implies that the system is primarily evaluated from the perspective of end users who directly interact with the research repository features. This distribution suggests that the usability results are strongly influenced by student experiences in managing research outputs and using system functionalities. For the course profile, BSCS respondents comprise 38.81 percent while both BSICT and BSIS each account for 30.60 percent, showing an almost even representation across computing-related programs. This balanced distribution across courses strengthens the validity of the evaluation since feedback comes from users with varied technical backgrounds. Regarding system experience, 66.42 percent of respondents have previously used a research repository system while 33.58 percent have not, indicating that most users have prior exposure which may positively influence their ability to assess system usability. Finally, in terms of research access frequency, the largest group accesses research materials weekly at 41.79 percent followed by daily users at 25.37 percent, while fewer respondents access monthly or rarely. This suggests that most participants are actively engaged in research activities, making them appropriate evaluators of SaliksikLab's functionality and usability.



Table 5. Mean, Standard Deviation, and Interpretation of the System Usability Scale (SUS) Results for the SaliksikLab System

Indicators	M	SD	VI	QD
I think that I would like to use this system frequently.	4.19	0.79	A	H
I found the system simple and not overly complex.	3.77	0.92	A	H
I thought the system was easy to use.	4.29	0.78	SA	VH
I think that I can use this system without needing assistance from a technical person.	3.68	0.70	A	H
I found the various functions in this system were well integrated.	4.06	0.77	A	H
I found the system consistent in its design and functionality.	3.84	0.82	A	H
I would imagine that most people would learn to use this system very quickly.	4.35	0.71	SA	VH
I found the system easy to navigate and not cumbersome.	3.94	0.77	A	H
I felt very confident using the system.	4.26	0.73	SA	H
I was able to use the system without needing to learn many things beforehand.	3.35	1.11	N	U
Overall Average:	3.97	0.81	A	H

The table 5 shows that the System Usability Scale (SUS) results of the SaliksikLab system indicate an overall positive usability perception with a mean score of 3.97 and a standard deviation of 0.81 interpreted as Agree with high usability. Most indicators received ratings ranging from Agree to Strongly Agree which suggests that users generally found the system easy to use, well integrated, and supportive of research management tasks. The highest-rated items include ease of use (M=4.29) and quick learnability (M=4.35) which indicates that users can adapt to the system with minimal difficulty. Respondents also expressed confidence in using the system (M=4.26) which reflects the effectiveness of the interface design and workflow structure. However, one indicator related to using the system without prior learning obtained a neutral response (M=3.35) which suggests that some users still require initial orientation before full utilization. The relatively low standard deviation across most items indicates consistent user perceptions with minimal variation in responses. Overall, the results imply that SaliksikLab provides a usable and accessible platform for research management within the academic setting.

The findings align with the principles of usability discussed by Nielsen [1] which emphasizes that a system should be easy to learn, efficient to use, and supportive of user confidence to ensure positive user experience. In addition, ISO/IEC 25010 usability standards highlight that software quality is strongly influenced by learnability, operability, and user satisfaction which are reflected in the positive SUS results of SaliksikLab. The high scores for ease of use and system integration suggest that the interface design effectively supports user tasks and reduces cognitive load during interaction. The presence of a neutral rating on initial learning requirements indicates that onboarding support or user guidance may further enhance usability. However, the overall strong agreement across most indicators confirms that the system meets acceptable usability standards for academic research platforms. The consistency of responses further strengthens the reliability of the usability findings. Therefore, the SUS results demonstrate that SaliksikLab is a user-friendly system that aligns with established usability frameworks in software design and evaluation.

Table 6. Target User Evaluation of SaliksikLab System

Indicators	M	SD	VI	QD
A. Functionality Suitability				
1. The system provides the necessary features for managing research outputs.	4.62	0.53	SA	VH
2. The repository effectively stores and organizes research documents.	4.64	0.51	SA	VH
3. The search and filtering functions help users locate research outputs efficiently.	4.60	0.54	SA	VH
4. The multi-language code execution feature supports software-based research projects effectively.	4.61	0.52	SA	VH
5. The analytics and reporting features provide useful information for monitoring research activities.	4.65	0.50	SA	VH
B. Performance Efficiency				
6. The system responds quickly to user actions.	4.63	0.52	SA	VH
7. Uploading and retrieving research documents is efficient.	4.62	0.53	SA	VH
8. The system performs well even with multiple users.	4.60	0.54	SA	VH

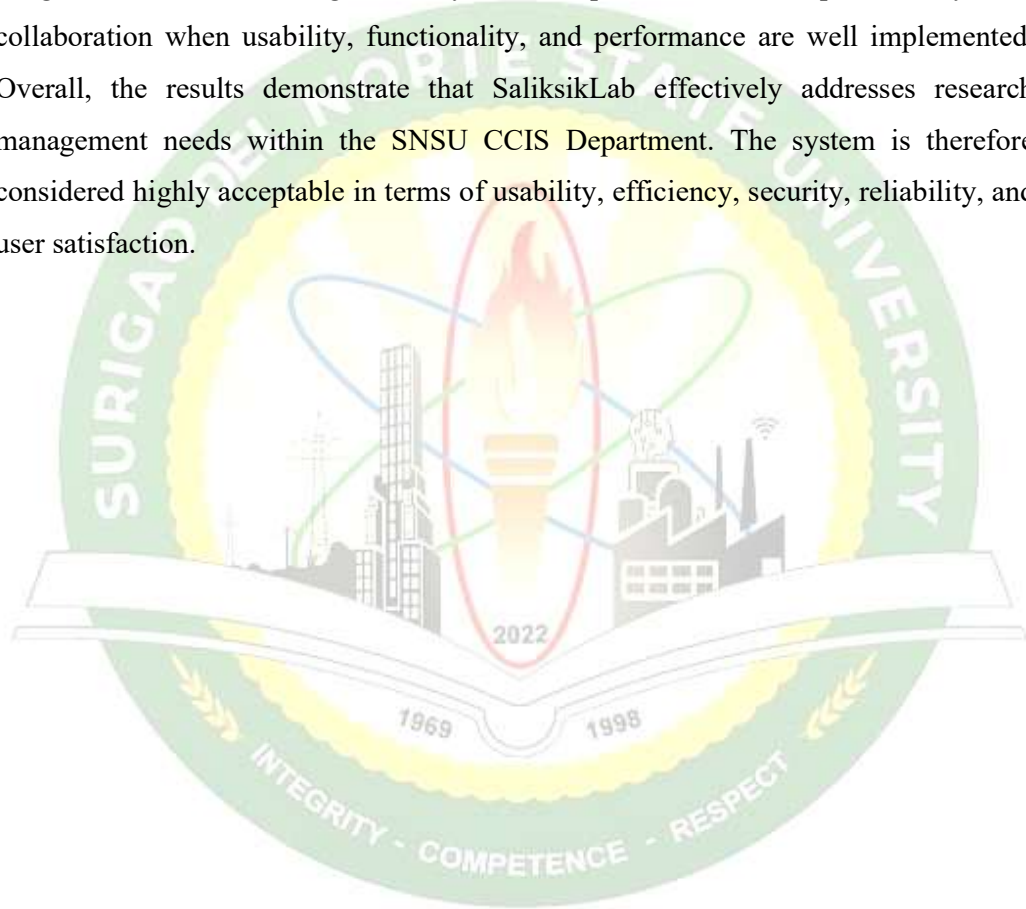
9. Code execution runs within an acceptable time.	4.58	0.56	SA	VH
10. The system uses resources efficiently without lag.	4.61	0.52	SA	VH
C. Security				
11. The system protects user accounts and information effectively.	4.64	0.51	SA	VH
12. The system restricts unauthorized access to research files and records.	4.63	0.52	SA	VH
13. The code execution environment appears safe and controlled.	4.61	0.53	SA	VH
14. The system maintains stable performance over time.	4.60	0.54	SA	VH
15. The system protects the privacy of uploaded research outputs.	4.65	0.50	SA	VH
D. Reliability				
16. The system operates consistently without crashes	4.55	0.56	SA	VH
17. The system performs tasks accurately	4.57	0.55	SA	VH
18. The system can recover from errors effectively	4.52	0.58	SA	VH
19. The system maintains stable performance over time	4.56	0.56	SA	VH
20. The system ensures data integrity during operations	4.58	0.55	SA	VH
E. User Satisfaction				
21. I am satisfied with the overall performance of the system.	4.74	0.47	SA	VH
22. The system improves access to academic research outputs.	4.76	0.45	SA	VH
23. The system makes research management more organized and efficient.	4.75	0.46	SA	VH
24. I would recommend the system to other students and instructors.	4.73	0.48	SA	VH
25. Overall, the system meets my expectations as a research management platform.	4.72	0.49	SA	VH
Grand Mean	4.64	0.52	SA	Very High

Scale	Interval	Verbal Interpretation	Code	Qualitative Description	Code
5	4.20-5.00	<i>Strongly Agree</i>	SA	<i>Very High</i>	VH
4	3.40-4.19	<i>Agree</i>	A	<i>High</i>	H
3	2.60-3.39	<i>Neutral</i>	N	<i>Undecided</i>	U
2	1.80-2.59	<i>Disagree</i>	D	<i>Low</i>	L
1	1.00-1.79	<i>Strongly Disagree</i>	SD	<i>Very Low</i>	VL

The table 6 shows that the target user evaluation of the SaliksikLab system during the testing phase obtained consistently high mean scores across all evaluated dimensions. The system achieved a grand mean of 4.64 with a standard deviation of 0.52, interpreted as Strongly Agree with a Very High qualitative description. This indicates that respondents generally perceived the system as highly effective in supporting research management activities. Across the indicators, Functionality Suitability received mean scores ranging from 4.60 to 4.65, suggesting that users strongly agreed that the system provides complete and appropriate features for research workflows. Performance Efficiency also yielded strong results, with means between 4.58 and 4.63, indicating that the system performs quickly and efficiently even under multi-user conditions. These results imply that SaliksikLab successfully meets both functional and operational requirements expected in a research repository system.

In terms of Security, the evaluation results show consistently high ratings with means ranging from 4.60 to 4.65, reflecting strong user confidence in data protection and access control mechanisms. Respondents strongly agreed that the system effectively protects user accounts, restricts unauthorized access, and ensures the privacy of uploaded research outputs. These findings suggest that the implementation of role-based access control and authentication mechanisms contributes significantly to system trustworthiness. Reliability indicators also produced high mean values from 4.52 to 4.58, indicating that the system operates consistently with minimal errors and maintains stable performance over time. Users also perceived that the system can recover from errors and maintain data integrity across modules. This supports the conclusion that SaliksikLab is stable and dependable for continuous academic use.

User Satisfaction received the highest overall ratings among all categories, with means ranging from 4.72 to 4.76, indicating very strong acceptance of the system. Respondents strongly agreed that SaliksikLab improves access to academic research outputs and enhances the organization and efficiency of research management processes. They also expressed willingness to recommend the system to other users, which reflects positive perceived value and usability in academic settings. The high satisfaction ratings align with findings from prior studies which emphasize that integrated research management systems improve academic productivity and collaboration when usability, functionality, and performance are well implemented. Overall, the results demonstrate that SaliksikLab effectively addresses research management needs within the SNSU CCIS Department. The system is therefore considered highly acceptable in terms of usability, efficiency, security, reliability, and user satisfaction.



Chapter 6

CONCLUSION AND RECOMMENDATIONS

6.1. Conclusion

This study aimed to design and develop SaliksikLab, a web-based research management system for the SNSU CCIS Department that improves the management, accessibility, security, and analysis of academic research outputs. The system integrates key features such as research repository management, role-based access control, secure multi-language code execution, and data analytics dashboards. The findings show that SaliksikLab effectively centralizes research documents and improves the organization, retrieval, and monitoring of academic outputs through structured metadata and search functionality. It also enhances academic workflows by allowing students and instructors to upload, review, and manage research outputs within a secure and centralized platform. The integration of a controlled code execution environment supports the validation of software-based research outputs while maintaining system security and stability. Furthermore, the analytics features provide meaningful insights that assist administrators in tracking research activity and institutional productivity.

The overall system evaluation results support these findings, with a grand mean of 4.64 indicating a Strongly Agree response and a Very High qualitative description. Respondents rated the system highly in terms of functional suitability, performance efficiency, security, reliability, usability, and user satisfaction. Usability results from the SUS also showed a high level of acceptance with an overall mean of 3.97, indicating that users found the system easy to use, well-integrated, and suitable for frequent use. These results imply that SaliksikLab is effective in addressing research management challenges within the university environment. However, limitations were observed in terms of long-term scalability testing and broader institutional deployment. Future improvements may focus on expanding system deployment scope, optimizing performance under larger datasets, and enhancing analytics capabilities for deeper research insights. Overall, SaliksikLab demonstrates that a well-designed research management system can significantly improve academic efficiency, collaboration, and research oversight in higher education institutions.

6.2. Recommendations

The researchers suggest several enhancements to improve the scalability, performance, and overall effectiveness of the SaliksikLab system as a research management platform. Based on the findings and limitations identified during development and evaluation, these recommendations aim to support future improvements and ensure the system remains adaptable to evolving academic needs. Enhancements should focus on strengthening system capabilities in code execution, data management, analytics, security, and user accessibility. Continuous improvement is also recommended to maintain system reliability and long-term usability within academic institutions. The following specific recommendations are provided for future researchers and developers:

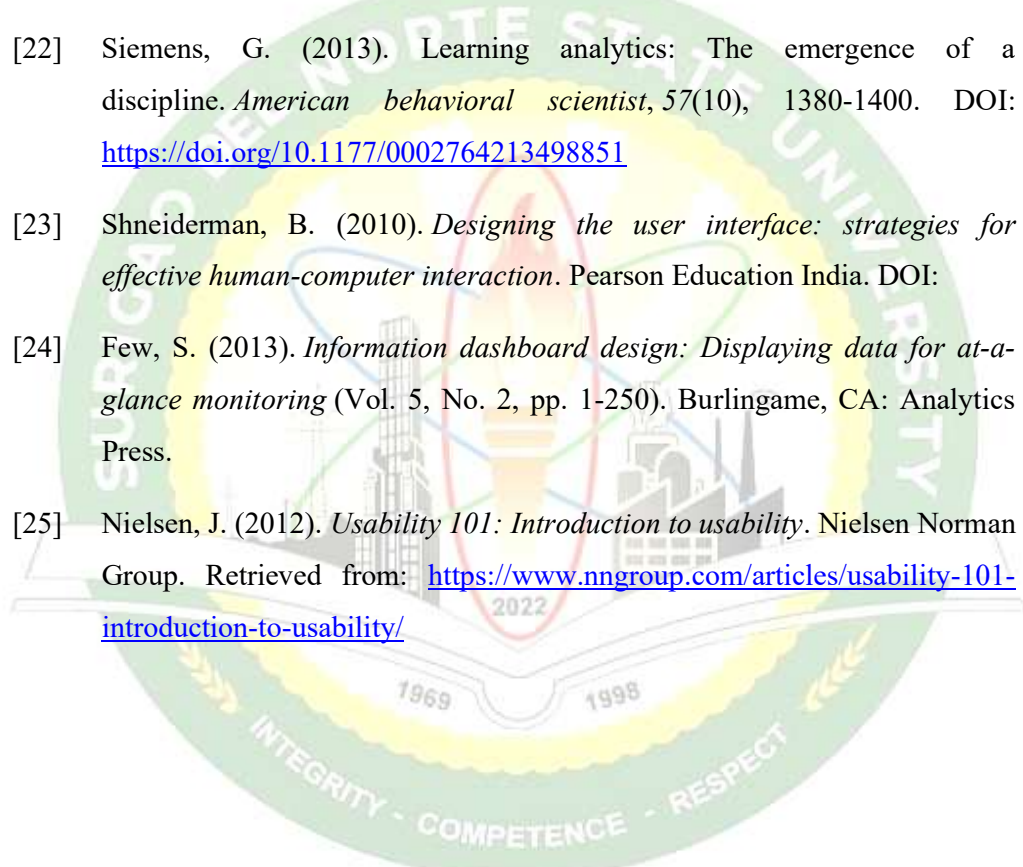
1. Integrate additional programming languages and enhance sandbox security mechanisms to ensure safer, more flexible, and more controlled code execution within the system.
2. Expand multilingual support to include additional languages to improve accessibility and support a more diverse academic user base.
3. Improve AI-assisted models for better accuracy in translation, summarization, and research content analysis to enhance data interpretation and usability.
4. Implement cloud-based infrastructure and distributed system architecture to improve scalability, system performance, and storage capacity for larger datasets and increased user demand.
5. Conduct comprehensive usability testing involving a wider range of users to further enhance system usability, interface design, and overall user experience.

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APPENDICES

Appendix A

LETTER ASKING PERMISSION TO FLOAT THE QUESTIONNAIRE/ CONDUCT INTERVIEW



Republic of the Philippines
SURIGAO DEL NORTE STATE UNIVERSITY
Narciso Street, Surigao City 8400, Philippines



February 18, 2026

DR. VIRNILLE C. FRANCISCO
Dean, College of Computing and Information Sciences
Surigao del Norte State University
Narciso St., Brgy. Taft, Surigao City

Dear Ma'am,

Good day!

We, the third-year Bachelor of Science in Computer Science (BSCS) students of Surigao del Norte State University (SNSU), are currently conducting our research entitled "Implementing SaliksikLab: An AI-Based Research Management System with Automated Content Analysis, Multilingual Processing, and Secure Multi-Language Code Execution" as partial fulfillment of the requirements for our course. The study aims to design and develop an intelligent research management system that enhances efficiency, organization, content analysis, and secure code execution to improve research productivity and collaboration within the institution.

As part of our data-gathering process, we respectfully request your permission to conduct interviews with selected faculty members and to administer questionnaires within the College of Computing and Information Sciences. The information gathered from the respondents will help us obtain valuable insights and will serve as a basis for designing a system that accurately addresses the needs of the college.

Thank you for your consideration. We look forward to your favorable response.

Sincerely,

JEOD LAINE E. SAEZ
BSCS Student


VINCE B. COMON
BSCS Student

Noted by:

UNIFE O. CAGAS, DTE
Thesis Professor/Adviser

Recommending Approval:

JESSICA ROSE E. FERNANDEZ, PhD
Faculty, CCIS

Approved by:

VIRNILLE C. FRANCISCO, PhD
Dean, CCIS

Tel. Nos.: (086) 827-3741

Email: info@snsu.edu.ph
URL: snsu.edu.ph

Appendix B

RESEARCH QUESTIONNAIRE

Title:

“Implementing SaliksikLab: A Research Management System with Secure Multi-Language Code Execution and Data Analytics for SNSU CCIS Department”

Direction:

Please answer all questions honestly. Your responses will be treated with the highest level of confidentiality and used for academic purposes only.

Part I – Demographic Profile

(Please check or fill in the appropriate answer.)

Name (Optional): _____

Age: _____

Sex: ☐ Male

☐ Female

Course:

☐ BSCS

☐ BSICT

☐ BSIS

User Role: ☐ Student ☐ Instructor

Have you used a research repository system before? ☐ Yes

☐ No

How often do you access research materials or repositories?

☐ Daily

☐ Weekly

☐ Monthly

☐ Rarely

Part II – System Usability

Direction: The following statements relate to the usability of the SaliksikLab system, including its ease of use, interface design, and overall user experience. Please check (✓) the column that best represents your level of agreement using the scale below.

Legend:

5 = Strongly Agree

4 = Agree

3 = Neutral

2 = Disagree

1 = Strongly Disagree

Statement	5 SA	4 A	3 N	2 D	1 SD
1. I think that I would like to use this system frequently.					
2. I found the system unnecessarily complex.					
3. I thought the system was easy to use.					
4. I think that I would need the support of a technical person to be able to use this system.					
5. I found the various functions in this system were well integrated.					

6. I thought there was too much inconsistency in this system.					
7. I would imagine that most people would learn to use this system very quickly.					
8. I found the system very cumbersome to use.					
9. I felt very confident using the system.					
10. I needed to learn a lot of things before I could get going with this system.					

Part III – System Quality Evaluation (ISO/IEC 25010)

Direction: The following statements relate to the evaluation of the SaliksikLab system based on ISO/IEC 25010 software quality standards, including functionality, performance, security, and reliability. Please check (✓) the column that best represents your level of agreement using the scale below.

Legend:

5 = Strongly Agree

4 = Agree

3 = Neutral

2 = Disagree

1 = Strongly Disagree

B. Functionality Suitability

Statement	5 SA	4 A	3 N	2 D	1 SD
1. The system provides the necessary features for managing research outputs.					
2. The repository effectively stores and organizes research documents.					
3. The search and filtering functions help users locate research outputs efficiently.					
4. The multi-language code execution feature supports software-based research projects effectively.					
5. The analytics and reporting features provide useful information for monitoring research activities.					

B. Performance Efficiency

Statement	5 SA	4 A	3 N	2 D	1 SD
6. The system responds quickly to user actions.					
7. Uploading and retrieving research documents is efficient.					
8. The system performs well even with multiple users.					
9. Code execution runs within an acceptable time.					
10. The system uses resources efficiently without lag.					

C. Security

Statement	5 SA	4 A	3 N	2 D	1 SD

11. The system protects user accounts and information effectively.					
12. The system restricts unauthorized access to research files and records.					
13. The code execution environment appears safe and controlled.					
14. The system maintains stable performance over time.					
15. The system protects the privacy of uploaded research outputs.					

D. Reliability

Statement	5 SA	4 A	3 N	2 D	1 SD
16. The system operates consistently without crashes					
17. The system performs tasks accurately					
18. The system can recover from errors effectively					
19. The system maintains stable performance over time					
20. The system ensures data integrity during operations					

E. User Satisfaction

Statement	5 SA	4 A	3 N	2 D	1 SD
21. I am satisfied with the overall performance of the system.					
22. The system improves access to academic research outputs.					
23. The system makes research management more organized and efficient.					
24. I would recommend the system to other students and instructors.					
25. Overall, the system meets my expectations as a research management platform.					

Direction: Please answer the following questions as thoughtfully and thoroughly as you can. Your responses will be treated with confidentiality and will only be used for research purposes.

1. What features of the SaliksikLab system did you find most useful in managing research outputs, and why?

Answer:

2. What challenges or difficulties did you encounter while using the system (e.g., navigation, code execution, search, or analytics)?

Answer:

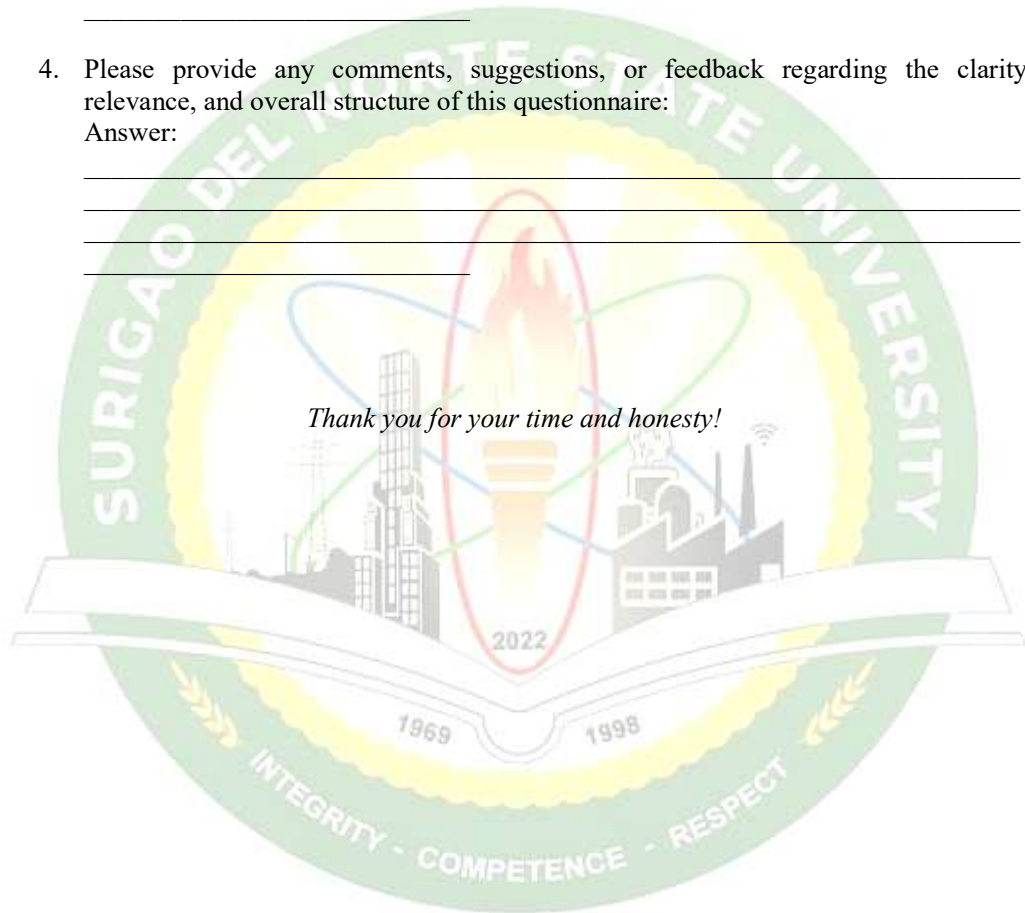
3. In what ways can the system be improved to enhance your overall user experience and support academic tasks more effectively?

Answer:

4. Please provide any comments, suggestions, or feedback regarding the clarity, relevance, and overall structure of this questionnaire:

Answer:

Thank you for your time and honesty!



Appendix C
CURRICULUM VITAE OF RESEARCHER

PERSONAL DATA

Name: Vince B. Comon
Email: v97742218@gmail.com
Contact No.: 09186428137
Date of Birth: July 17, 2004
Religion: Roman Catholic
Nationality: Filipino
Address: Purok 1, Barangay Katipunan, Del Carmen, SDN



EDUCATIONAL BACKGROUND

Tertiary:

Undergraduate A.Y.: 2023 - 2027
Surigao del Norte State University
Narciso St., Surigao City, Surigao del Norte

Secondary:

High School A.Y.: 2017 - 2023
Surigao del Norte State University – Del Camen Campus
Purok 6, San Jose, Del Carmen, Siargao Island, Surigao del Norte

Researcher Position:

Web Developer
Tester
Technical Writer

Appendix C
CURRICULUM VITAE OF RESEARCHER

PERSONAL DATA

Name: Jeody Laine E. Saez
Email: saezjeodylaine@gmail.com
Contact No.: 09514512770
Date of Birth: June 5, 2005
Religion: IFI
Nationality: Filipino
Address: Purok 5, By. Bonifacio, Surigao City, SDN



EDUCATIONAL BACKGROUND

Tertiary:

Undergraduate A.Y.: 2023 - 2027
Surigao del Norte State University
Narciso St., Surigao City, Surigao del Norte

Secondary:

Senior High School A.Y.: 2021 - 2023
Caraga Regional Science High School
Roland Sering St., Purok 10, Brgy San Juan, Surigao City, Surigao del Norte

Junior High School A.Y.: 2017 - 2021
Northeastern Mindanao Colleges
Amat Corner, Rizal Street, Surigao City, Surigao del Norte

Researcher Position:

UI/UX Designer
Database Engineer
Technical Writer